

Particles and Nuclei 2

- 3/3/26
- Announcements:
- Schedule : Take home midterm/ HW 4 combo next

- Takehome problems work independently
- Others you can collaborate on

Wk 9 3/17	
Wk 10 3/24	Take hom Midtrm / HW 4 due
Wk 11 3/31	Friday HW 5 due
Wk 12 4/7	
Wk 13 4/14	HW 6 Due
Wk 14 4/21	
Finals week	Final Project

- For next time Read Griff Ch 7.1, 7.2 ~7.3ish
 - Dirac Equation!

Appreciate these first ?

- $1 \rightarrow 2+3$ Decay COM

$$\Gamma = \frac{S|\mathbf{p}|}{8\pi\hbar m_1^2 c} |\mathcal{M}|^2$$

more generally

$$\frac{1}{\tau} = \Gamma = \frac{|\vec{p}^*|}{32\pi^2 m_i^2} \int |\mathcal{M}_{fi}|^2 d\Omega$$

- Now we see a scale of how much the kinematics (phase space) matters in the decays e.g. statements in earlier chapters...

- QUANTITATIVELY!!!!

Simplified versions: Our Master Formulae

- $1 \rightarrow 2+3$ Decay COM

$$\Gamma = \frac{S|\mathbf{p}|}{8\pi\hbar m_1^2 c} |\mathcal{M}|^2$$

Griff's

more generally

$$\frac{1}{\tau} = \Gamma = \frac{|\vec{p}^*|}{32\pi^2 m_i^2} \int |M_{fi}|^2 d\Omega$$

Thomson

- Cross section $1+2 \rightarrow 3+4$

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar c}{8\pi}\right)^2 \frac{S|\mathcal{M}|^2}{(E_1 + E_2)^2} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}$$

$$\sigma = \frac{1}{64\pi^2 s} \frac{|\vec{p}_f^*|}{|\vec{p}_i^*|} \int |M_{fi}|^2 d\Omega^*$$

Better invariant
version

$$\frac{d\sigma}{dt} = \frac{1}{64\pi s |\vec{p}_i^*|^2} |M_{fi}|^2$$

Exercise (last time)

- For a decay $1 \rightarrow 2 + 2$
- (meaning single particle mass m_1 decays to 2 identical particles with m_2)
- derive an expression for $|p|$, either final state momentum in the COM frame in terms of the masses of the two particles

$$|p| = \sqrt{\left(\frac{m_1}{2}\right)^2 - m_2^2}$$

$$\sqrt{\frac{s}{4} - m_2^2}$$

Example

$$m_{D^0} = 1.85 \text{ GeV}$$

$$m/2 = 0.925 \text{ MeV} \sqrt{.92^2 - 0.138^2} \approx 0.92$$

(for pi basically all $m/2$ goes to momentum
but not too much less in case of K)

$$\frac{\Gamma(D^0 \rightarrow \pi^- \pi^+)}{\Gamma(D^0 \rightarrow K^- \pi^+)} = \frac{|V_{cd}|^2}{|V_{cs}|^2} \left(\frac{p_{\pi^-}}{p_{K^-}} \right) \approx \lambda^2 \left(\frac{922}{861} \right)$$

e.g. kinematic enhancement factor is 1.07 but dynamical suppression factor is $\lambda^2 \sim .05$.

$$D^0 \rightarrow \pi^+ \pi^-$$

• Branching Fraction: $1.455 \pm 0.025 \times 10^{-3}$.

• Branching Fraction: $B(D^0 \rightarrow K^- \pi^+) \approx 3.95 \times 10^{-2}$.

- TAKE RATIO $\Gamma_{\pi}/\Gamma_K \rightarrow$ everything cancels but $\mathcal{M}$ and $|p|$; hope $\mathcal{M}$ doesn't matter $\rightarrow$ this case CKM's it does much more than phase space

How did we get simplified versions?

$$\delta^4(p) \equiv \delta^4(p^\mu) = \delta(p_0) \delta(p_x) \delta(p_y) \delta(p_z)$$

Example same massless(or same) particles for 2+3

Consider decays of the form $1 \rightarrow 2+3+ \dots + N$

$$d\Gamma = |\mathcal{M}|^2 \frac{S}{2\hbar m_1} \left[\left(\frac{cd^3\vec{p}_2}{(2\pi)^3 2E_2} \right) \left(\frac{cd^3\vec{p}_3}{(2\pi)^3 2E_3} \right) \dots \left(\frac{cd^3\vec{p}_n}{(2\pi)^3 2E_n} \right) \right] \times (2\pi)^4 \delta^4(p_1 - p_2 - \dots - p_n)$$

$$\Gamma = \frac{S}{\hbar m} \left(\frac{1}{4\pi} \right)^2 \frac{1}{2} \int \frac{|\mathcal{M}|^2}{|\vec{p}_2| |\vec{p}_3|} \delta(mc - |\vec{p}_2| - |\vec{p}_3|) \delta^3(-\vec{p}_2 - \vec{p}_3) d^3\vec{p}_2 d^3\vec{p}_3$$

$\underbrace{|\vec{p}_2|}_{E_2} = \underbrace{|\vec{p}_3|}_{E_3}$

$\underbrace{E_2, E_3}_{E_{2,3}}$

1 → 2+3 Decay (massless 2,3)

$$\Gamma = \frac{S}{\hbar m} \left(\frac{1}{4\pi} \right)^2 \frac{1}{2} \int \frac{|\mathcal{M}|^2}{|\vec{p}_2| |\vec{p}_3|} \delta(mc - |\vec{p}_2| - |\vec{p}_3|) \delta^3(-\vec{p}_2 - \vec{p}_3) d^3\vec{p}_2 d^3\vec{p}_3$$

$\underbrace{|\vec{p}_2|}_{E_2} = \underbrace{|\vec{p}_3|}_{E_3}$ $\underbrace{|\vec{p}_2| + |\vec{p}_3|}_{E_{2,3}}$ $\nwarrow$

$$\Gamma = \frac{S}{2(4\pi)^2 \hbar m} \int \frac{|\mathcal{M}|^2}{|\vec{p}_2|} \delta(mc - 2|\vec{p}_2|) d^3\vec{p}_2 \quad \mathcal{M} = \mathcal{M}(|\vec{p}_2|)$$

$\underbrace{d^3\vec{p}_2}_{r^2 dr d\Omega}$

$$d^3\vec{p}_2 = |\vec{p}_2|^2 d|\vec{p}_2| \sin\theta d\theta d\phi$$

Do angular integral

$$\int d\Omega = 4\pi$$

$$\delta(k_x) = \frac{1}{k} \delta(x)$$

$$\Gamma = \frac{S}{8\pi\hbar m} \int_0^\infty |\mathcal{M}|^2 \frac{1}{2} \delta\left(|\vec{p}_2| - \frac{mc}{2}\right) d|\vec{p}_2|$$

$$\Gamma = \frac{S}{16\pi\hbar m} |\mathcal{M}|^2$$

$$\rightarrow \mathcal{M}\left(\frac{m}{2}\right)$$

1 → 2+3 Decay Massive $m_2 = m_3$

$$\Gamma = \frac{S}{\hbar m} \left(\frac{1}{4\pi} \right)^2 \frac{1}{2} \int \frac{|\mathcal{M}|^2}{|\vec{p}_2| |\vec{p}_3|} \delta(mc - |\vec{p}_2| - |\vec{p}_3|) \delta^3(-\vec{p}_2 - \vec{p}_3) d^3\vec{p}_2 d^3\vec{p}_3$$

$E_2 = E_3$ $E_{2,3}$ $(\times 4\pi)$

$$\Gamma = \frac{S}{2(4\pi)^2 \hbar m} \int \frac{|\mathcal{M}|^2}{|\vec{p}_2|^2} \delta(mc - 2|\vec{p}_2|) d^3\vec{p}_2 \rightarrow [] \int \frac{M^2}{E_2} \delta(mc - 2E_2) p_2^2 dp_2$$

$$\frac{dE_2}{dp_2} = \frac{p_2}{E_2} \Rightarrow \int M^2 \delta(mc - 2E_2) \frac{p_2}{E_2} dE_2$$

$\swarrow$ $m_{1/2}$

$$\delta(kx) = \frac{1}{k} \delta(x)$$

$$\Gamma = \frac{S |\mathbf{p}|}{8\pi \hbar m_1^2 c} |\mathcal{M}|^2$$

p_2

$$p_2 = \sqrt{E_2^2 - m_2^2} = \sqrt{\left(\frac{m_1}{2}\right)^2 - m_2^2}$$

from last δ fn

Case 2 $J/\Psi \rightarrow 2\text{lepton}$

- $1 \rightarrow 2+3$ Decay COM

$$\Gamma = \frac{S|\mathbf{p}|}{8\pi\hbar m_1^2 c} |\mathcal{M}|^2$$

more generally

$$\frac{1}{\tau} = \Gamma = \frac{|\vec{p}^*|}{32\pi^2 m_i^2} \int |M_{fi}|^2 d\Omega$$

- Scale of 4-momentum phase space effects

J/Ψ ($m \sim 3\text{GeV}$) $\rightarrow \mu\mu$ or to $ee \rightarrow |p|$ same both cases:
 $\sqrt{1.5^2 - 0^2}$ $\sqrt{s}$ $\sqrt{1.5^2 - 0.1^2}$

(both m small compared to 1.5 GeV) rate is about the same (1% level) – works well here $\rightarrow$ less other things diff?!? $c\bar{c}$ annihilation $\rightarrow$ EM Photon or Z

$J/\psi \rightarrow \mu^+ \mu^-$ Branching Ratio:

- **Branching Fraction:** $5.961\% \pm 0.033\%$ or $(5.960 \pm 0.065 \pm 0.050)\%$.
- **Comparison to e^+e^- :** The ratio $B(J/\psi \rightarrow e^+e^-)/B(J/\psi \rightarrow \mu^+\mu^-)$ is $(99.7 \pm 1.2 \pm 0.6)\%$.

Case 3 $B \rightarrow p$ vs $\rightarrow \Lambda$

- $1 \rightarrow 2+3$ Decay COM

$$\Gamma = \frac{S|\mathbf{p}|}{8\pi\hbar m_1^2 c} |\mathcal{M}|^2$$

more generally

$$\frac{1}{\tau} = \Gamma = \frac{|\vec{p}^*|}{32\pi^2 m_i^2} \int |M_{fi}|^2 d\Omega$$

- Scale of 4-momentum phase space effects
- **Example 3 :** First, Bmeson decay $m = 5 \text{ GeV}$
 - $E_i = m_B / 2$ $p_i = \sqrt{E_i^2 - m_i^2} \rightarrow$ pions $\rightarrow |p| = 2.5 \text{ GeV}$
 - Proton/Lambda ($m_\Lambda = 1.1 \text{ GeV}$) $= |p| = \sqrt{2.5^2 - 1} = 2.3 \text{ GeV}$
 - For Delta(1.3 GeV)/Lambda(1.1) $|p| = \sqrt{2.5^2 - 1} \sim 2.0 \text{ GeV}$

le the Kinematics difference should only reduce by $\sim 10\%$ or 20% not a factor of 10 less! --- *doesn't work very well again (right direction though), matrix element must matter more!*

Light unflavored meson modes				
$\rho^+ \gamma$	$(9.8 \pm 2.5) \times 10^{-7}$		2583	
$\pi^+ \pi^0$	$(5.5 \pm 0.4) \times 10^{-6}$	S=1.2	2636	
$p \bar{\Lambda}$	$(2.4 \pm \frac{1.0}{0.9}) \times 10^{-7}$		2430	
$\Delta^+ \bar{\Lambda}$	< 8.2	$\times 10^{-7}$	CL=90%	—

Decay Rate Calculations

- For the two body decay

$$i \rightarrow 1 + 2$$

$$\begin{aligned} M_{fi} &= \langle \psi'_1 \psi'_2 | \hat{H}' | \psi_i \rangle \\ &= (2E_i \cdot 2E_1 \cdot 2E_2)^{1/2} \langle \psi_1 \psi_2 | \hat{H}' | \psi_i \rangle \\ &= (2E_i \cdot 2E_1 \cdot 2E_2)^{1/2} T_{fi} \end{aligned}$$

★ Now expressing T_{fi} in terms of M_{fi} gives

$$\Gamma_{fi} = \frac{(2\pi)^4}{2E_i} \int |M_{fi}|^2 \delta(E_i - E_1 - E_2) \delta^3(\vec{p}_i - \vec{p}_1 - \vec{p}_2) \frac{d^3 \vec{p}_1}{(2\pi)^3 2E_1} \frac{d^3 \vec{p}_2}{(2\pi)^3 2E_2}$$

Note:

- M_{fi} uses relativistically normalised wave-functions. It is Lorentz Invariant
- $\frac{d^3 \vec{p}}{(2\pi)^3 2E}$ is the Lorentz Invariant Phase Space for each final state particle
the factor of $2E$ arises from the wave-function normalisation
(prove this in Question 2)
- This form of Γ_{fi} is simply a rearrangement of the original equation
but the integral is now frame independent (i.e. L.I.)
- Γ_{fi} is inversely proportional to E_i , the energy of the decaying particle. This is exactly what one would expect from time dilation ($E_i = \gamma m$).
- Energy and momentum conservation in the delta functions

Decay Rate Calculations

$$\Gamma_{fi} = \frac{(2\pi)^4}{2E_i} \int |M_{fi}|^2 \delta(E_i - E_1 - E_2) \delta^3(\vec{p}_i - \vec{p}_1 - \vec{p}_2) \frac{d^3 \vec{p}_1}{(2\pi)^3 2E_1} \frac{d^3 \vec{p}_2}{(2\pi)^3 2E_2}$$

★ Because the **integral** is Lorentz invariant (i.e. frame independent) it can be evaluated in any frame we choose. The C.o.M. frame is most convenient

- In the C.o.M. frame $E_i = m_i$ and $\vec{p}_i = 0$ 

$$\Gamma_{fi} = \frac{1}{8\pi^2 E_i} \int |M_{fi}|^2 \delta(m_i - E_1 - E_2) \delta^3(\vec{p}_1 + \vec{p}_2) \frac{d^3 \vec{p}_1}{2E_1} \frac{d^3 \vec{p}_2}{2E_2}$$

- Integrating over $\vec{p}_2$ using the δ -function:



$$\Gamma_{fi} = \frac{1}{8\pi^2 E_i} \int |M_{fi}|^2 \delta(m_i - E_1 - E_2) \frac{d^3 \vec{p}_1}{4E_1 E_2}$$

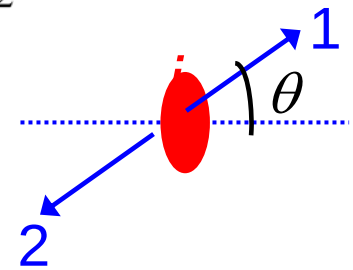
now $E_2^2 = (m_2^2 + |\vec{p}_1|^2)$ since the δ -function imposes $\vec{p}_2 = -\vec{p}_1$

- Writing $d^3 \vec{p}_1 = p_1^2 dp_1 \sin \theta d\theta d\phi = p_1^2 dp_1 d\Omega$

For convenience, here $|\vec{p}_1|$ is written as p_1



$$\Gamma_{fi} = \frac{1}{32\pi^2 E_i} \int |M_{fi}|^2 \delta \left(m_i - \sqrt{m_1^2 + p_1^2} - \sqrt{m_2^2 + p_1^2} \right) \frac{p_1^2 dp_1 d\Omega}{E_1 E_2}$$

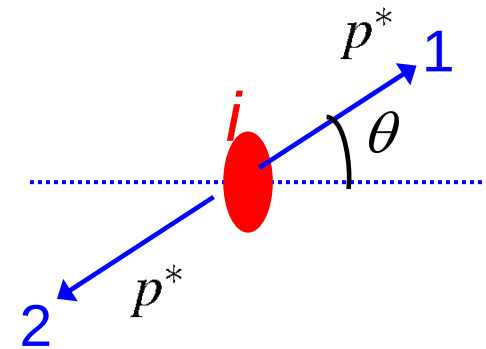


- Which can be written in the form
- $$\Gamma_{fi} = \frac{1}{32\pi^2 E_i} \int |M_{fi}|^2 g(p_1) \delta(f(p_1)) dp_1 d\Omega \quad (2)$$

where $g(p_1) = p_1^2 / (E_1 E_2) = p_1^2 (m_1^2 + p_1^2)^{-1/2} (m_2^2 + p_1^2)^{-1/2}$

and $f(p_1) = m_i - (m_1^2 + p_1^2)^{1/2} - (m_2^2 + p_1^2)^{1/2}$

- Note:
- $\delta(f(p_1))$ imposes energy conservation.
 - $f(p_1) = 0$ determines the C.o.M momenta of the two decay products
i.e. $f(p_1) = 0$ for $p_1 = p^*$



- ★ Eq. (2) can be integrated using the property of δ -function derived earlier (eq. (1))

$$\int g(p_1) \delta(f(p_1)) dp_1 = \frac{1}{|df/dp_1|_{p^*}} \int g(p_1) \delta(p_1 - p^*) dp_1 = \frac{g(p^*)}{|df/dp_1|_{p^*}}$$

where p^* is the value for which $f(p^*) = 0$

- All that remains is to evaluate df/dp_1

$$\frac{df}{dp_1} = -\frac{p_1}{(m_1^2 + p_1^2)^{1/2}} - \frac{p_1}{(m_2^2 + p_1^2)^{1/2}} = -\frac{p_1}{E_1} - \frac{p_1}{E_2} = -p_1 \frac{E_1 + E_2}{E_1 E_2}$$

giving:

$$\Gamma_{fi} = \frac{1}{32\pi^2 E_i} \int |M_{fi}|^2 \left| \frac{E_1 E_2}{p_1 (E_1 + E_2)} \frac{p_1^2}{E_1 E_2} \right|_{p_1=p^*} d\Omega$$

$$= \frac{1}{32\pi^2 E_i} \int |M_{fi}|^2 \left| \frac{p_1}{E_1 + E_2} \right|_{p_1=p^*} d\Omega$$

- But from $f(p_1) = 0$, i.e. energy conservation: $E_1 + E_2 = m_i$

$$\Gamma_{fi} = \frac{|\vec{p}^*|}{32\pi^2 E_i m_i} \int |M_{fi}|^2 d\Omega$$

In the particle's rest frame $E_i = m_i$



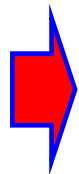
$$\left[\frac{1}{\tau} = \Gamma = \frac{|\vec{p}^*|}{32\pi^2 m_i^2} \int |M_{fi}|^2 d\Omega \right] \quad (3)$$

VALID FOR ALL TWO-BODY DECAYS !

- p^* can be obtained from $f(p_1) = 0$

$$(m_1^2 + p^{*2})^{1/2} + (m_2^2 + p^{*2})^{1/2} = m_i$$

(Question 3)



$$p^* = \frac{1}{2m_i} \sqrt{[(m_i^2 - (m_1 + m_2)^2) [m_i^2 - (m_1 - m_2)^2]]} \quad \text{(now try Questions 4 \& 5)}$$

Simplified versions: Our Master Formulae

- $1 \rightarrow 2+3$ Decay COM

$$\Gamma = \frac{S|\mathbf{p}|}{8\pi\hbar m_1^2 c} |\mathcal{M}|^2$$

Griff's

more generally

$$\frac{1}{\tau} = \Gamma = \frac{|\vec{p}^*|}{32\pi^2 m_i^2} \int |M_{fi}|^2 d\Omega$$

Thomson

- Cross section $1+2 \rightarrow 3+4$

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar c}{8\pi}\right)^2 \frac{S|\mathcal{M}|^2}{(E_1 + E_2)^2} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}$$

$$\sigma = \frac{1}{64\pi^2 s} \frac{|\vec{p}_f^*|}{|\vec{p}_i^*|} \int |M_{fi}|^2 d\Omega^*$$

Better invariant
version

$$\frac{d\sigma}{dt} = \frac{1}{64\pi s |\vec{p}_i^*|^2} |M_{fi}|^2$$

Simplifications for Cross Section/ Similar

- Probably won't go through similar simplification in Griffiths for $1+2 \rightarrow 3+4$ Scattering (?)

For the process $1 + 2 \rightarrow 3 + 4 + \dots + N$

$$d\sigma = |\mathcal{M}|^2 \frac{\hbar^2 S}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}} \left[\left(\frac{cd^3 \vec{p}_3}{(2\pi)^3 2E_3} \right) \left(\frac{cd^3 \vec{p}_4}{(2\pi)^3 2E_4} \right) \dots \left(\frac{cd^3 \vec{p}_n}{(2\pi)^3 2E_n} \right) \right] \times (2\pi)^4 \delta^4(p_1 + p_2 - \dots - p_n)$$

- But $\times \prod_{j=3}^n 2\pi \delta(p_j^2 - m_j^2 c^2) \theta(p_j^0) \frac{d^4 p_j}{(2\pi)^4}$ *whole big thing just $\equiv d^3 p/E$* t that process

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar c}{8\pi} \right)^2 \frac{S |\mathcal{M}|^2}{(E_1 + E_2)^2} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}$$

- Now there are more integrals, less delta functions – how do we again are we able to avoid integrating over the M^2 ?
- Answer: Instead of doing the angular integrals, move the angular part (or more generally can just move one of the whole $d^3 p/E$ for invariant x.s.)

$$d\sigma = |\mathcal{M}|^2 \frac{\hbar^2 S}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}} \left[\left(\frac{cd^3 \vec{p}_3}{(2\pi)^3 2E_3} \right) \right]$$

$p_3^2 d p_3 d\Omega$

$\frac{d\sigma}{d\Omega} \quad E d\sigma$

$\frac{d\sigma}{d^3 p_3}$

$\left(\frac{d\sigma}{E d p_3 d\Omega} \right)$

could be 4 (or 5, 6, 7...)

Flux

The scattering Golden Rule has this non-trivial looking factor “FLUX”

For the process $1 + 2 \rightarrow 3 + 4 + \dots + N$

$$d\sigma = |\mathcal{M}|^2 \frac{\hbar^2 S}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}} \left[\left(\frac{cd^3 \vec{p}_3}{(2\pi)^3 2E_3} \right) \left(\frac{cd^3 \vec{p}_4}{(2\pi)^3 2E_4} \right) \dots \left(\frac{cd^3 \vec{p}_n}{(2\pi)^3 2E_n} \right) \right] \times (2\pi)^4 \delta^4(p_1 + p_2 - \dots - p_n)$$

From Thomson's (Roche) slides we see a nice couple cases of this factor written in different perhaps more recognizable forms. These are useful to use along with the formula above itself.

$$\sigma = \frac{(2\pi)^{-2}}{2E_1 2E_2 (v_1 + v_2)} \int |M_{fi}|^2 \delta(E_1 + E_2 - E_3 - E_4) \delta^3(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4) \frac{d^3 \vec{p}_3}{2E_3} \frac{d^3 \vec{p}_4}{2E_4}$$

- The **integral** is now written in a **Lorentz invariant** form
- The quantity $F = 2E_1 2E_2 (v_1 + v_2)$ can be written in terms of a four-vector scalar product and is therefore also **Lorentz Invariant** (the Lorentz Inv. Flux)

$$F = 4 [(p_1^\mu p_{2\mu})^2 - m_1^2 m_2^2]^{1/2} \quad (\text{see appendix I})$$

- Consequently cross section is a **Lorentz Invariant** quantity

Two special cases of Lorentz Invariant Flux:

- **Centre-of-Mass Frame**

$$\begin{aligned} F &= 4E_1 E_2 (v_1 + v_2) \\ &= 4E_1 E_2 (|\vec{p}^*|/E_1 + |\vec{p}^*|/E_2) \\ &= 4|\vec{p}^*|(E_1 + E_2) \\ &= 4|\vec{p}^*|\sqrt{s} \end{aligned}$$

- **Target (particle 2) at rest**

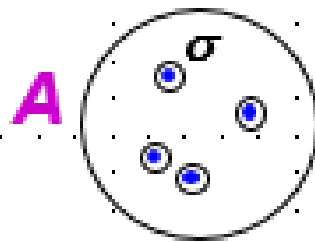
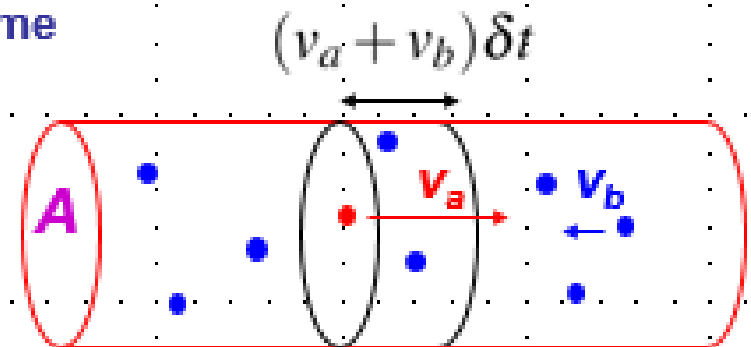
$$\begin{aligned} F &= 4E_1 E_2 (v_1 + v_2) \\ &= 4E_1 m_2 v_1 \\ &= 4E_1 m_2 (|\vec{p}_1|/E_1) \\ &= 4m_2 |\vec{p}_1| \end{aligned}$$

Flux

Thomson slides

- Consider a single particle of type **a** with velocity, $\mathbf{v}_a$, traversing a region of area **A** containing n_b particles of type **b** per unit volume

In time δt a particle of type **a** traverses region containing $n_b(v_a + v_b)A\delta t$ particles of type **b**



- ★ Interaction probability obtained from effective cross-sectional area occupied by the $n_b(v_a + v_b)A\delta t$ particles of type **b**

Interaction Probability =
$$\frac{n_b(v_a + v_b)A\delta t\sigma}{A} = n_b v \delta t \sigma \quad [v = v_a + v_b]$$



Rate per particle of type **a** = $n_b v \sigma$

- Consider volume **V**, total reaction rate = $(n_b v \sigma) \cdot (n_a V) = (n_b V) (n_a v) \sigma$
 $= N_b \phi_a \sigma$

Flux

The scattering Golden Rule has this non-trivial looking factor “FLUX”

$$\sigma = \frac{(2\pi)^{-2}}{2E_1 2E_2 (v_1 + v_2)} \int |M_{fi}|^2 \delta(E_1 + E_2 - E_3 - E_4) \delta^3(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4) \frac{d^3\vec{p}_3}{2E_3} \frac{d^3\vec{p}_4}{2E_4}$$

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$$F = 4 \left[(p_1^\mu p_{2\mu})^2 - m_1^2 m_2^2 \right]^{1/2} \quad (\text{see appendix I})$$

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$$\begin{aligned} F &= 4E_1 E_2 (v_1 + v_2) \\ &= 4E_1 E_2 (|\vec{p}^*|/E_1 + |\vec{p}^*|/E_2) \\ &= 4|\vec{p}^*|(E_1 + E_2) \\ &= 4|\vec{p}^*|\sqrt{s} \end{aligned}$$

- **Target (particle 2) at rest**

$$\begin{aligned} F &= 4E_1 E_2 (v_1 + v_2) \\ &= 4E_1 m_2 v_1 \\ &= 4E_1 m_2 (|\vec{p}_1|/E_1) \\ &= 4m_2 |\vec{p}_1| \end{aligned}$$

Flux

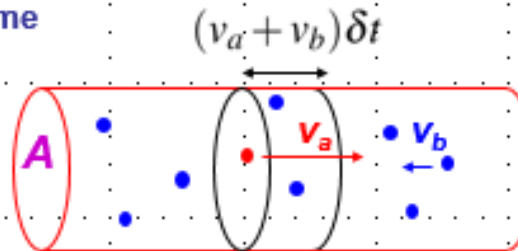
For the process $1 + 2 \rightarrow 3 + 4 + \dots + N$

$$d\sigma = |\mathcal{M}|^2 \frac{\hbar^2 S}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}} \left[\left(\frac{cd^3 \vec{p}_3}{(2\pi)^3 2E_3} \right) \left(\frac{cd^3 \vec{p}_4}{(2\pi)^3 2E_4} \right) \dots \left(\frac{cd^3 \vec{p}_n}{(2\pi)^3 2E_n} \right) \right] \times (2\pi)^4 \delta^4(p_1 + p_2 - \dots - p_n)$$

The reason it is literally from "beam" flux – see here::

- Consider a single particle of type **a** with velocity, $\mathbf{v}_a$, traversing a region of area **A** containing n_b particles of type **b** per unit volume

In time δt a particle of type **a** traverses region containing $n_b(v_a + v_b)A\delta t$ particles of type **b**



★ Interaction probability obtained from effective cross-sectional area occupied by the $n_b(v_a + v_b)A\delta t$ particles of type **b**

• Interaction Probability = $\frac{n_b(v_a + v_b)A\delta t\sigma}{A} = n_b v \delta t \sigma$ $[v = v_a + v_b]$



Rate per particle of type **a** = $n_b v \sigma$

• Consider volume **V**, total reaction rate = $(n_b v \sigma) \cdot (n_a V) = (n_b V) (n_a v) \sigma$
 $= N_b \phi_a \sigma$

Interesting point:
 $v_a + v_b$ does not need relativistic addition here – it can be $2c$! b.c. the rate of crossings in our frame is not limited by the S.R. constraint, $v_a + v_b$ even for both $v_i \sim c$

Flux

- $p^2/m \propto vE$

$$\begin{aligned}\text{Rate/Volume} &= (\text{flux of 1}) \times (\text{number density of 2}) \times \sigma \\ &= n_1(v_1 + v_2) \times n_2 \times \sigma\end{aligned}$$

Simplified versions – x.s. Appreciated

- Cross section $1+2 \rightarrow 3+4 \rightarrow$ SPINLESS

Better invariant version

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar c}{8\pi}\right)^2 \frac{S|\mathcal{M}|^2}{(E_1 + E_2)^2} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}$$

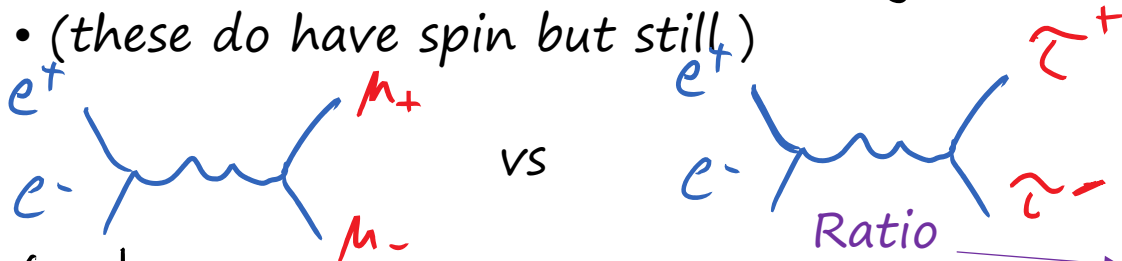
$$\frac{d\sigma}{dt} = \frac{1}{64\pi s |\vec{p}_i^*|^2} |M_{fi}|^2$$

- (in COM)

- Note, first also kinematic effect

- A given process with similar matrix elements (similar physics, like both simple QED) is more likely by factor of final state momenta – from kine only

- (these do have spin but still)



$$m_\mu = 0.1 \text{ GeV}$$

$$m_\tau = 2 \text{ GeV}$$

to be checked expect
x.s./rate of $\mu\mu \sim 2\times$
higher than $\tau\tau$

$$|\mathbf{p}| = \sqrt{s/4 - m_i^2} \quad \text{pick } s = 5 \text{ GeV}$$

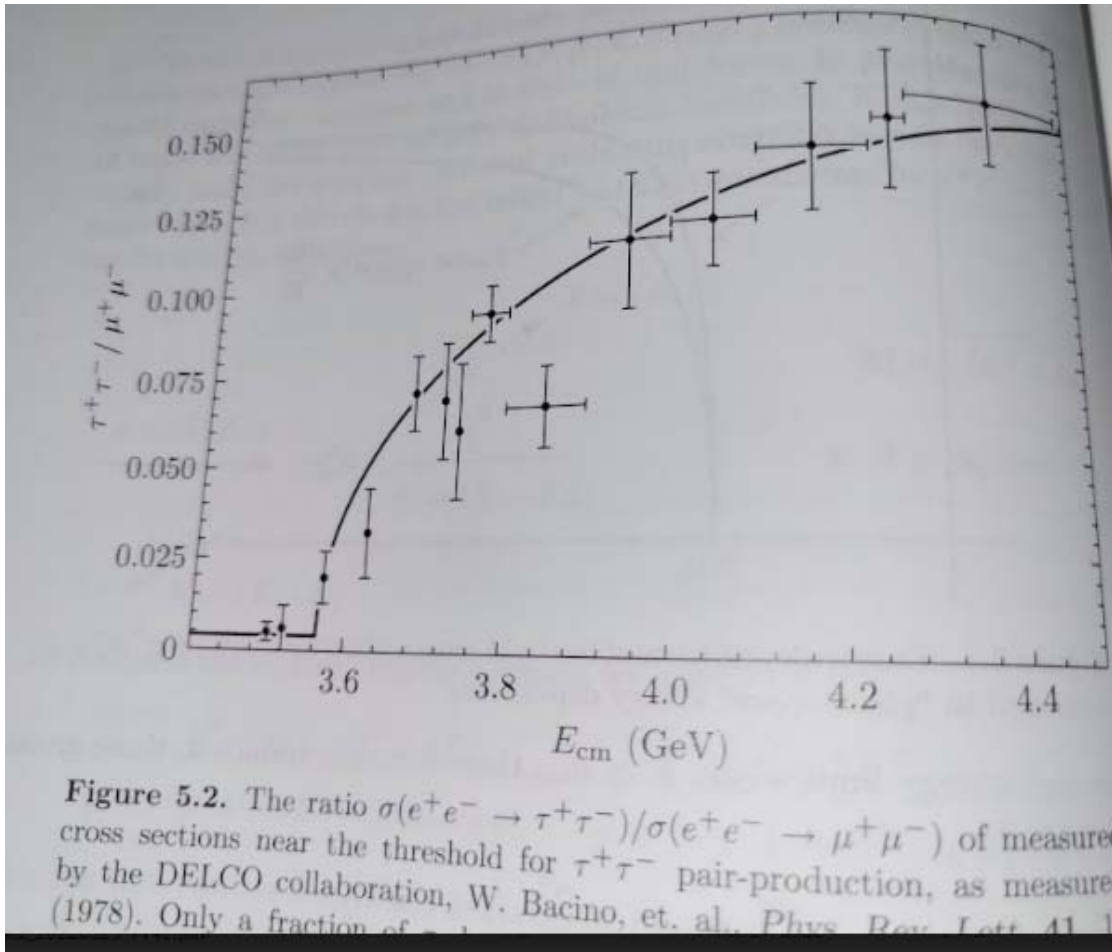
Ratio
 $p_{f\mu}/p_{f\tau}$

$$\frac{2.5 \text{ GeV} (e)}{\sqrt{25/4 - 4} = \sqrt{2}}$$

$$\frac{2.5}{1.4}$$

Not bad

- ... in Peskin



our previous estimate is the same order of magnitude

I'd say that's as good as could be expected we wanted to see approx how much the kinematic phases space affects the cross section by itself

Simplified versions

- $1 \rightarrow 2+3$ Decay COM

$$\Gamma = \frac{S|\mathbf{p}|}{8\pi\hbar m_1^2 c} |\mathcal{M}|^2$$

more generally

$$\frac{1}{\tau} = \Gamma = \frac{|\vec{p}^*|}{32\pi^2 m_i^2} \int |M_{fi}|^2 d\Omega$$

- Cross section $1+2 \rightarrow 3+4$

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar c}{8\pi}\right)^2 \frac{S|\mathcal{M}|^2}{(E_1 + E_2)^2} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}$$

$$\sigma = \frac{1}{64\pi^2 s} \frac{|\vec{p}_f^*|}{|\vec{p}_i^*|} \int |M_{fi}|^2 d\Omega^*$$

Better invariant
version

$$\frac{d\sigma}{dt} = \frac{1}{64\pi s |\vec{p}_i^*|^2} |M_{fi}|^2$$

Feynman Rules QFT for M

- Now we learn simple Feynman rules for getting matrix elements themselves

For the process $1 + 2 \rightarrow 3 + 4 + \dots + N$

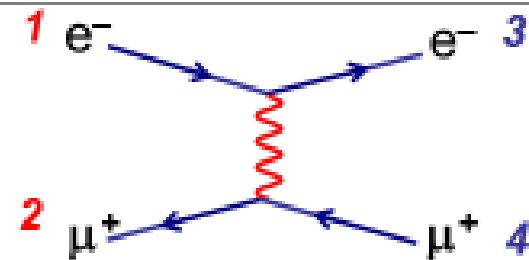
$$d\sigma = |\mathcal{M}|^2 \frac{\hbar^2 S}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}} \left[\left(\frac{cd^3\vec{p}_3}{(2\pi)^3 2E_3} \right) \left(\frac{cd^3\vec{p}_4}{(2\pi)^3 2E_4} \right) \dots \left(\frac{cd^3\vec{p}_n}{(2\pi)^3 2E_n} \right) \right] \times (2\pi)^4 \delta^4(p_1 + p_2 - \dots - p_n)$$

- It will generally get integrated over

Invariant x.s. expression

- In the case of elastic scattering $|\vec{p}_i^*| = |\vec{p}_f^*|$

$$\sigma_{\text{elastic}} = \frac{1}{64\pi^2 s} \int |M_{fi}|^2 d\Omega^*$$

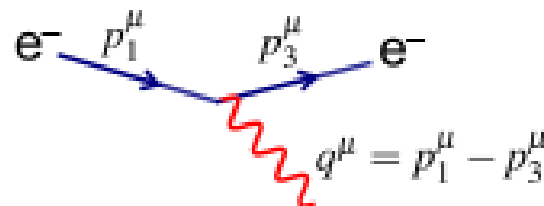


- For calculating the total cross-section (which is Lorentz Invariant) the result on the previous page (eq. (4)) is sufficient. However, it is not so useful for calculating the differential cross section in a rest frame other than the C.o.M:

$$d\sigma = \frac{1}{64\pi^2 s} \frac{|\vec{p}_f^*|}{|\vec{p}_i^*|} |M_{fi}|^2 d\Omega^*$$

because the angles in $d\Omega^* = d(\cos \theta^*) d\phi^*$ refer to the C.o.M frame

- For the last calculation in this section, we need to find a L.I. expression for $d\sigma$
- ★ Start by expressing $d\Omega^*$ in terms of Mandelstam t i.e. the square of the four-momentum transfer



$$t = q^2 = (p_1 - p_3)^2$$

Product of four-vectors therefore L.I.

Invariant x.s. formula

- Want to express $d\Omega^*$ in terms of Lorentz Invariant dt

where $t \equiv (p_1 - p_3)^2 = p_1^2 + p_3^2 - 2p_1 \cdot p_3 = m_1^2 + m_3^2 - 2p_1 \cdot p_3$

- In C.o.M. frame:

$$p_1^{*\mu} = (E_1^*, 0, 0, |\vec{p}_1^*|)$$

$$p_3^{*\mu} = (E_3^*, |\vec{p}_3^*| \sin \theta^*, 0, |\vec{p}_3^*| \cos \theta^*)$$

$$p_1^\mu p_{3\mu} = E_1^* E_3^* - |\vec{p}_1^*| |\vec{p}_3^*| \cos \theta^*$$

$$t = m_1^2 + m_3^2 - E_1^* E_3^* + 2|\vec{p}_1^*| |\vec{p}_3^*| \cos \theta^*$$

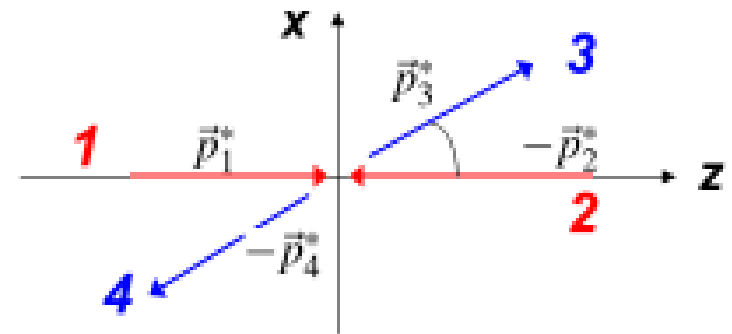
giving $dt = 2|\vec{p}_1^*| |\vec{p}_3^*| d(\cos \theta^*)$

therefore $d\Omega^* = d(\cos \theta^*) d\phi^* = \frac{dt d\phi^*}{2|\vec{p}_1^*| |\vec{p}_3^*|}$

hence $d\sigma = \frac{1}{64\pi^2 s} \frac{|\vec{p}_3^*|}{|\vec{p}_1^*|} |M_{fi}|^2 d\Omega^* = \frac{1}{2 \cdot 64\pi^2 s |\vec{p}_1^*|^2} |M_{fi}|^2 d\phi^* dt$

- Finally, integrating over $d\phi^*$ (assuming no ϕ^* dependence of $|M_{fi}|^2$) gives:

$$\frac{d\sigma}{dt} = \frac{1}{64\pi s |\vec{p}_i^*|^2} |M_{fi}|^2$$



Variations of Inv. x.s. expression

Lorentz Invariant differential cross section

- All quantities in the expression for $d\sigma/dt$ are Lorentz Invariant and therefore, it applies to **any rest frame**. It should be noted that $|\vec{p}_i^*|^2$ is a constant, fixed by energy/momentum conservation

$$|\vec{p}_i^*|^2 = \frac{1}{4s} [s - (m_1 + m_2)^2] [s - (m_1 - m_2)^2]$$

- As an example of how to use the invariant expression $d\sigma/dt$ we will consider elastic scattering in the laboratory frame in the limit where we can neglect the mass of the incoming particle $E_1 \gg m_1$

E_1 m_2 e.g. electron or neutrino scattering

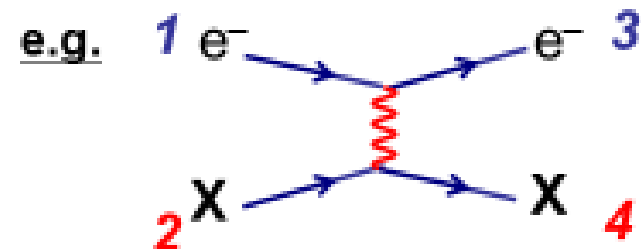
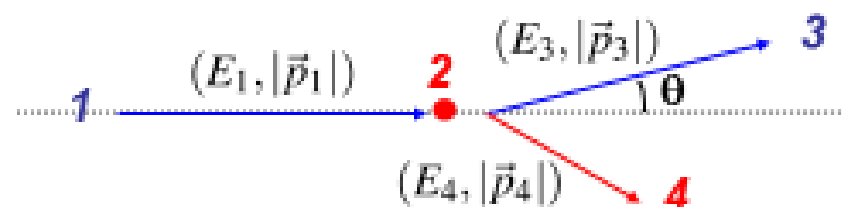
In this limit $|\vec{p}_i^*|^2 = \frac{(s - m_2^2)^2}{4s}$

$$\boxed{\frac{d\sigma}{dt} = \frac{1}{16\pi(s - m_2^2)^2} |M_{fi}|^2} \quad (m_1 = 0)$$

$d\sigma/dt$

2→2 Body Scattering in Lab. Frame

- The other commonly occurring case is scattering from a fixed target in the Laboratory Frame (e.g. electron-proton scattering)
- First take the case of elastic scattering at high energy where the mass of the incoming particles can be neglected: $m_1 = m_3 = 0$, $m_2 = m_4 = M$



- Wish to express the cross section in terms of scattering angle of the e^-

$$\text{therefore} \quad \frac{d\sigma}{d\Omega} = \frac{d\sigma}{dt} \frac{dt}{d\Omega} = \frac{1}{2\pi} \frac{dt}{d(\cos \theta)} \frac{d\sigma}{dt}$$

Integrating over $d\phi$

- The rest is some rather tedious algebra.... start from four-momenta

$$p_1 = (E_1, 0, 0, E_1), \quad p_2 = (M, 0, 0, 0), \quad p_3 = (E_3, E_3 \sin \theta, 0, E_3 \cos \theta), \quad p_4 = (E_4, \vec{p}_4)$$

$$\text{so here} \quad t = (p_1 - p_3)^2 = -2p_1 \cdot p_3 = -2E_1 E_3 (1 - \cos \theta)$$

But from (E,p) conservation $p_1 + p_2 = p_3 + p_4$

and, therefore, can also express t in terms of particles 2 and 4

Using $d\sigma/dt \rightarrow d\sigma/d\Omega$

$$\begin{aligned} t &= (p_2 - p_4)^2 = 2M^2 - 2p_2 \cdot p_4 = 2M^2 - 2ME_4 \\ &= 2M^2 - 2M(E_1 + M - E_3) = -2M(E_1 - E_3) \end{aligned}$$

Note E_1 is a constant (the energy of the incoming particle) so

$$\frac{dt}{d(\cos \theta)} = 2M \frac{dE_3}{d(\cos \theta)}$$

• Equating the two expressions for t gives

$$E_3 = \frac{E_1 M}{M + E_1 - E_1 \cos \theta}$$

so
$$\frac{dE_3}{d(\cos \theta)} = \frac{E_1 M}{(M + E_1 - E_1 \cos \theta)^2} = E_1^2 M \left(\frac{E_3}{E_1 M} \right)^2 = \frac{E_3^2}{M}$$

$$\frac{d\sigma}{d\Omega} = \frac{1}{2\pi} \frac{dt}{d(\cos \theta)} \frac{d\sigma}{dt} = \frac{1}{2\pi} 2M \frac{E_3^2}{M} \frac{d\sigma}{dt} = \frac{E_3^2}{\pi} \frac{d\sigma}{dt} = \frac{E_3^2}{\pi} \frac{1}{16\pi(s - M^2)^2} |M_{fi}|^2$$

using $s = (p_1 + p_2)^2 = M^2 + 2p_1 \cdot p_2 = M^2 + 2ME_1$
gives $(s - M^2) = 2ME_1$

Particle 1 massless
→ $(p_1^2 = 0)$

$$\rightarrow \frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2} \left(\frac{E_3}{ME_1} \right)^2 |M_{fi}|^2$$

In limit $m_1 \rightarrow 0$

further versions

In this equation, E_3 is a function of θ :

$$E_3 = \frac{E_1 M}{M + E_1 - E_1 \cos \theta}$$

giving

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2} \left(\frac{1}{M + E_1 - E_1 \cos \theta} \right)^2 |M_{fi}|^2 \quad (m_1 = 0)$$

General form for 2→2 Body Scattering in Lab. Frame

★ The calculation of the differential cross section for the case where m_1 can not be neglected is longer and contains no more “physics” (see appendix II). It gives:

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2} \cdot \frac{1}{p_1 m_1} \cdot \frac{|\vec{p}_3|^2}{|\vec{p}_3|(E_1 + m_2) - E_3 |\vec{p}_1| \cos \theta} \cdot |M_{fi}|^2$$

Again there is only one independent variable, θ , which can be seen from conservation of energy

$$E_1 + m_2 = \sqrt{|\vec{p}_3|^2 + m_3^2} + \sqrt{|\vec{p}_1|^2 + |\vec{p}_3|^2 - 2|\vec{p}_1||\vec{p}_3|\cos\theta + m_4^2}$$

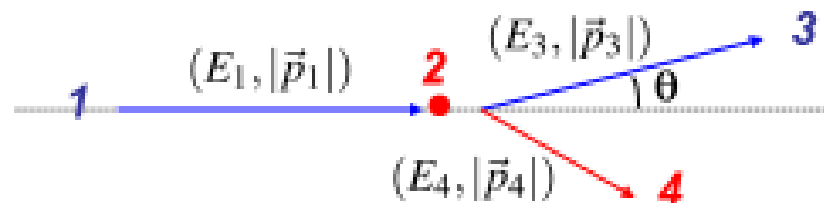
i.e. $|\vec{p}_3|$ is a function of θ

$$\vec{p}_4 = \vec{p}_1 - \vec{p}_3$$

Similar example using $d\sigma/dt$

Appendix II : general 2→2 Body Scattering in lab frame

NON-EXAMINABLE



$$p_1 = (E_1, 0, 0, |\vec{p}_1|), \quad p_2 = (M, 0, 0, 0), \quad p_3 = (E_3, E_3 \sin \theta, 0, E_3 \cos \theta), \quad p_4 = (E_4, \vec{p}_4)$$

again

$$\frac{d\sigma}{d\Omega} = \frac{d\sigma}{dt} \frac{dt}{d\Omega} = \frac{1}{2\pi} \frac{dt}{d(\cos \theta)} \frac{d\sigma}{dt}$$

But now the invariant quantity t :

$$\begin{aligned} t &= (p_2 - p_4)^2 = m_2^2 + m_4^2 - 2p_2 \cdot p_4 = m_2^2 + m_4^2 - 2m_2 E_4 \\ &= m_2^2 + m_4^2 - 2m_2 (E_1 + m_2 - E_3) \end{aligned}$$

$$\rightarrow \frac{dt}{d(\cos \theta)} = 2m_2 \frac{dE_3}{d(\cos \theta)}$$

...using $d\sigma/dt$ (cont)

Which gives
$$\frac{d\sigma}{d\Omega} = \frac{m_2}{\pi} \frac{dE_3}{d(\cos \theta)} \frac{d\sigma}{dt}$$

To determine $dE_3/d(\cos \theta)$, first differentiate $E_3^2 - |\vec{p}_3|^2 = m_3^2$

$$2E_3 \frac{dE_3}{d(\cos \theta)} = 2|\vec{p}_3| \frac{d|\vec{p}_3|}{d(\cos \theta)} \quad (\text{All.1})$$

Then equate $t = (p_1 - p_3)^2 = (p_4 - p_2)^2$ to give

$$m_1^2 + m_3^2 - 2(E_1 E_3 - |\vec{p}_1| |\vec{p}_3| \cos \theta) = m_4^2 + m_2^2 - 2m_2(E_1 + m_2 - E_3)$$

Differentiate wrt. $\cos \theta$

$$(E_1 + m_2) \frac{dE_3}{d \cos \theta} - |\vec{p}_1| \cos \theta \frac{d|\vec{p}_3|}{d \cos \theta} = |\vec{p}_1| |\vec{p}_3|$$

Using (1) $\rightarrow$
$$\frac{dE_3}{d(\cos \theta)} = \frac{|\vec{p}_1| |\vec{p}_3|^2}{|\vec{p}_3| (E_1 + m_2) - E_3 |\vec{p}_1| \cos \theta} \quad (\text{All.2})$$

$$\frac{d\sigma}{d\Omega} = \frac{m_2}{\pi} \frac{dE_3}{d(\cos \theta)} \frac{d\sigma}{dt} = \frac{m_2}{\pi} \frac{dE_3}{d(\cos \theta)} \frac{1}{64\pi s |\vec{p}_i^*|^2} |M_{fi}|^2$$

cont

It is easy to show $|\vec{p}_i^*| \sqrt{s} = m_2 |\vec{p}_1|$

$$\frac{d\sigma}{d\Omega} = \frac{dE_3}{d(\cos \theta)} \frac{m_2}{64\pi^2 m_2^2 |\vec{p}_1|^2} |M_{fi}|^2$$

and using (All.2) obtain

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2} \cdot \frac{1}{p_1 m_1} \cdot \frac{|\vec{p}_3|^2}{|\vec{p}_3|(E_1 + m_2) - E_3 |\vec{p}_1| \cos \theta} \cdot |M_{fi}|^2$$

Now for Matrix elements

- Using Feynman rules for Matrix elements in combo w/ these formulas, especially for cross section

Example Toy (ABC) Theory Feynman Rules

• Feynman Rules to Calculate M

Our problem is to find the amplitude $\mathcal{M}$ associated with a given Feynman diagram. The ritual is as follows [5]:

1. **Notation:** Label the incoming and outgoing four-momenta $p_1, p_2, \dots, p_n$ (Figure 6.6). Label the internal momenta $q_1, q_2, \dots$. Put an arrow beside each line, to keep track of the 'positive' direction (forward in time for external lines, arbitrary for internal lines).
2. **Vertex factors:** For each vertex, write down a factor

$$-ig$$

g is called the *coupling constant*; it specifies the strength of the interaction between A, B, and C. In this toy theory, g has the dimensions of momentum; in the 'real-world' theories, we shall encounter later on, the coupling constant is always dimensionless.

3. **Propagators:** For each internal line, write a factor

$$\frac{i}{q_j^2 - m_j^2 c^2}$$

where q_j is the four-momentum of the line and m_j is the mass of the particle the line describes. (Note that $q_j^2 \neq m_j^2 c^2$, because a virtual particle does not lie on its mass shell.)

4. **Conservation of energy and momentum:** For each vertex, write a delta function of the form

$$(2\pi)^4 \delta^4(k_1 + k_2 + k_3)$$

where the k 's are the three four-momenta coming into the vertex (if the arrow leads outward, then k is minus the four-momentum of that line). This factor imposes conservation of energy and momentum at each vertex since the delta function is zero unless the sum of the incoming momenta equals the sum of the outgoing momenta.

5. **Integration over internal momenta:** For each internal line, write down a factor^a

$$\frac{1}{(2\pi)^4} d^4 q_j$$

and integrate over all internal momenta.

6. **Cancel the delta function:** The result will include a delta function

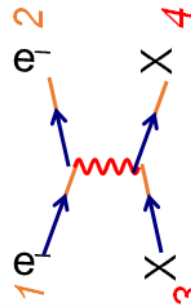
$$(2\pi)^4 \delta^4(p_1 + p_2 + \dots + p_n)$$

reflecting overall conservation of energy and momentum. Erase this factor^b and multiply by i . The result is $\mathcal{M}$.

Summarized

- 1) Diagram – Notate (1,2,3,etc)
- 2) Each Vertex factor F – coupling g
 $\rightarrow F = -ig$
- 3) (Each) propagator $i/(q^2 - m^2)$
- 4) Eac vertex ("uvw") $\delta(p_u \pm p_v \pm p_w)$ ie 4v mom conservation 3legs/arrows
- 5) Integrate internal lines $d^4 q$
- 6) Cancel/erase the final delta fn (δ^4)
 --global 4v mom consv
- 7) Multiply by i

tree diagram δ fn's: 2 (8) δ fns [two 4v] – 1(4) Integrals – 1 (4) final = no "real" integrations!



Particles and Nuclei 2

- 3/5/26
- Announcements:
- Schedule : Take home midterm/ HW 4 combo next

- Takehome problems work independently
- Others you can collaborate on

Wk 9 3/17	
Wk 10 3/24	Take hom Midtrm / HW 4 due
Wk 11 3/31	Friday HW 5 due
Wk 12 4/7	
Wk 13 4/14	HW 6 Due
Wk 14 4/21	
Finals week	Final Project

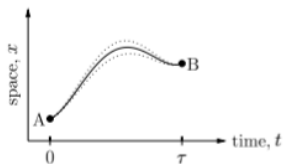
- Over break :Read Rest of Griff Ch 7 and possibly other readings will be specified

Outline

- *Griff Chapt 6* → *Feynman Rules / Toy Model*
 - *Some alternatives*
- *Start Dirac Equation Ch 7*

Note : QFT “Feynman Propagators”/Lagrangian

- OldFashPertbTh (OFPT) derivation of “propagator”/Feynman rules used standard <Hint> PT
- Hamiltonian Formulation also still used in both FT books for main method of introduction.
- Lagrangian ONLY used to define formally the H. Sorry you still won't really get to see “Feynman Path Integrals” in action except as NOTATION



The integral of the Lagrangian over time is known as the action S

$$S = \int_0^\tau L dt, \quad (1.25)$$

$$\frac{\delta S}{\delta x(t)} = 0, \quad (1.26)$$

and this is known as **Hamilton's principle of least action**.² It states that the classical trajectory taken by a particle is such that the action is stationary: small adjustments to the path taken by the particle (Fig. 1.4)

“Path Integral” but really doesn't use Lagrangian, just wave fn's $\langle x|x' \rangle = \int \langle x|p \rangle \langle p|x' \rangle$

Free part ($\delta(x/p)$)

One way to do quantum mechanics is to calculate a wave function and operate on it with quantum operators. Another way is to directly consider amplitudes for a given process, such as ‘the amplitude that my particle starts at point y at a time t_y and ends up at point x at time t_x ’, which we write down¹ as $\langle x(t_x)|y(t_y) \rangle$. This amplitude is known as a **propagator**. We've spent time in previous chapters removing the

16

Propagators and Green's functions

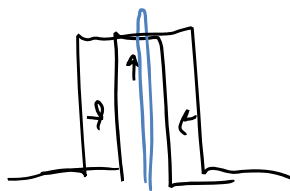
- True “least action” path integral w/ Lagrangian not used except for “more powerful” re-formulations of QFT using Functional Theory introduced way later in books, & not used much there –probably for a theorist just provides 2nd check of methods – overblown!!!

Representations of $\delta(x)$

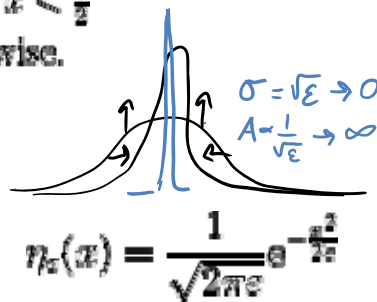
- Limiting cases of many functions: (wikipedia convention)

$$\delta(x) = \lim_{\varepsilon \rightarrow 0^+} \eta_\varepsilon(x),$$

$$\eta_\varepsilon(x) = \frac{1}{\varepsilon} \text{rect}\left(\frac{x}{\varepsilon}\right) = \begin{cases} \frac{1}{\varepsilon}, & -\frac{\varepsilon}{2} < x < \frac{\varepsilon}{2} \\ 0, & \text{otherwise.} \end{cases}$$



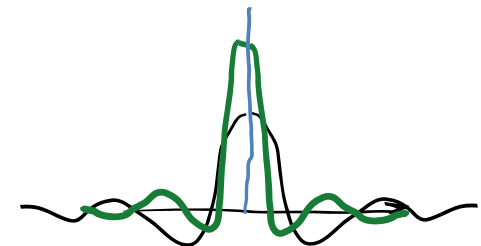
Rect fn



Gaussian

$$\eta_\varepsilon(x) = \frac{1}{\pi} \frac{\varepsilon}{\varepsilon^2 + x^2}$$

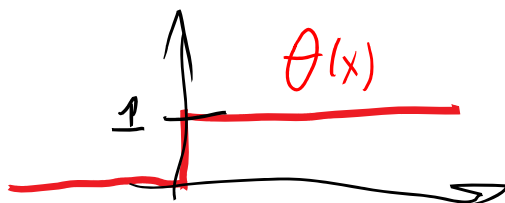
Lorentzian



$$\eta_\varepsilon(x) = \frac{1}{\varepsilon} \frac{\sin(x/\varepsilon)}{\pi x/\varepsilon} = \frac{1}{\varepsilon} \text{sinc}\left(\frac{x}{\varepsilon}\right)$$

sinc x

- Two VERY important other ones:



Step Fn / $\theta(x)$

$$\delta(x) = \theta'(x) = \frac{d\theta}{dx} \quad \text{Griff hw prob}$$

$$\theta(x) = \frac{d?}{dx} = R' \quad R''(x) = \delta(x)$$



$$\delta(x) = \lim_{L \rightarrow \infty} \frac{1}{2\pi} \int_{-L}^L dk e^{ikx}$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} dk e^{ikx} !$$

"Exponential Integral"

Other

- $\delta(x) = 1/(x+i\epsilon)$

Wikipedia: Propagators

- *Relativistic* : this derivation explains some of the previous steps better

The position space propagators are [Green's functions](#) for the [Klein–Gordon equation](#). This means that they are functions $G(x, y)$ satisfying

$$(\square_x + m^2)G(x, y) = -\delta(x - y),$$

where

x, y are two points in [Minkowski spacetime](#),

$\square_x = \frac{\partial^2}{\partial t^2} - \nabla^2$ is the [d'Alembertian](#) operator acting on the x coordinates,

$\delta(x - y)$ is the [Dirac delta-function](#).

(As typical in [relativistic](#) quantum field theory calculations, we use units where the [speed of light](#) c and [Planck's reduced constant](#) $\hbar$ are set to unity.)

We shall restrict attention to 4-dimensional [Minkowski spacetime](#). We can perform a [Fourier transform](#) of the equation for the propagator, obtaining

$$(-p^2 + m^2)G(p) = -1.$$

This equation can be inverted in the sense of [distributions](#), noting that the equation $xf(x) = 1$ has the solution (see [Sokhotski–Plemelj theorem](#))

$$f(x) = \frac{1}{x \pm i\varepsilon} = \frac{1}{x} \mp i\pi\delta(x),$$

with ε implying the limit to zero. Below, we discuss the right choice of the sign arising from causality requirements.

The solution is

$$G(x, y) = \frac{1}{(2\pi)^4} \int d^4p \frac{e^{-ip(x-y)}}{p^2 - m^2 \pm i\varepsilon},$$

Ignore $i\varepsilon$ it is just the delta function for E conservation

Propagator

- Outline some parts of L & B

The position space propagators are [Green's functions](#) for the [Klein–Gordon equation](#). This means that they are functions $G(x, y)$ satisfying

$$(\square_x + m^2)G(x, y) = -\delta(x - y),$$

where

x, y are two points in [Minkowski spacetime](#),

$\square_x = \frac{\partial^2}{\partial t^2} - \nabla^2$ is the [d'Alembertian](#) operator acting on the x coordinates,

$\delta(x - y)$ is the [Dirac delta-function](#).

(As typical in [relativistic](#) quantum field theory calculations, we use units where the [speed of light](#) c and [Planck's reduced constant](#) $\hbar$ are set to unity.)

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$$G(x, y) = \frac{1}{(2\pi)^4} \int d^4p \frac{e^{-ip(x-y)}}{p^2 - m^2 \pm i\varepsilon},$$

Redux

$$\begin{aligned} G(x, y) &= \langle \Omega | T \hat{\phi}(x) \hat{\phi}^\dagger(y) | \Omega \rangle \\ &= \theta(x^0 - y^0) \langle \Omega | \hat{\phi}(x) \hat{\phi}^\dagger(y) | \Omega \rangle + \theta(y^0 - x^0) \langle \Omega | \hat{\phi}(y)^\dagger \hat{\phi}(x) | \Omega \rangle, \end{aligned}$$

$$\begin{aligned} [\Delta(x, y)]_{(2)} &\equiv \theta(y^0 - x^0) \langle 0 | \hat{\phi}^\dagger(y) \hat{\phi}(x) | 0 \rangle \\ &= \theta(y^0 - x^0) \int \frac{d^3 p}{(2\pi)^3 (2E_p)} e^{-iE_p(y^0 - x^0) + i\mathbf{p} \cdot (\mathbf{y} - \mathbf{x})} \\ &= i \int_{-\infty}^{\infty} \frac{dz d^3 p}{(2\pi)^4 (2E_p)} \frac{e^{-i(E_p + z)(y^0 - x^0) + i\mathbf{p} \cdot (\mathbf{y} - \mathbf{x})}}{z + i\epsilon}. \end{aligned} \quad (17.20)$$

The substitution is now $z' = z + E_p$, giving

$$[\Delta(x, y)]_{(2)} = i \int_{-\infty}^{\infty} \frac{dz' d^3 p}{(2\pi)^4 (2E_p)} \frac{e^{-iz'(y^0 - x^0) + i\mathbf{p} \cdot (\mathbf{y} - \mathbf{x})}}{p^0 - E_p + i\epsilon}. \quad (17.21)$$

Lastly, make the substitution $p \rightarrow -p$, which is to say $(p^0, \mathbf{p}) \rightarrow (-p^0, -\mathbf{p})$, we get

$$\begin{aligned} [\Delta(x, y)]_{(2)} &= -i \int_{-\infty}^{\infty} \frac{dz' d^3 p}{(2\pi)^4 (2E_p)} \frac{e^{-ip^0(x^0 - y^0) + i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})}}{p^0 + E_p - i\epsilon} \\ &= -i \int_{-\infty}^{\infty} \frac{d^4 p}{(2\pi)^4 (2E_p)} \frac{e^{-ip \cdot (x - y)}}{p^0 + E_p - i\epsilon}. \end{aligned} \quad (17.22)$$

The full propagator $\Delta(x, y) = [\Delta(x, y)]_{(1)} + [\Delta(x, y)]_{(2)}$ is, therefore, the sum of eqns 17.19 and 17.22:

$$\begin{aligned} \Delta(x, y) &= i \int \frac{d^4 p}{(2\pi)^4 (2E_p)} e^{-ip \cdot (x - y)} \left(\frac{1}{p^0 - E_p + i\epsilon} - \frac{1}{p^0 + E_p - i\epsilon} \right) \\ &= \int \frac{d^4 p}{(2\pi)^4} e^{-ip \cdot (x - y)} \frac{i}{(p^0)^2 - E_p^2 + i\epsilon}. \end{aligned} \quad (17.23)$$

$$G(x, y) = \frac{1}{(2\pi)^4} \int d^4 p \frac{e^{-ip \cdot (x - y)}}{p^2 - m^2 \pm i\epsilon},$$

$$f(x) = \frac{1}{x \pm i\epsilon} =$$

with ϵ implying the limit

The solution is

Redux

$$\begin{aligned} G(x, y) &= \langle \Omega | T \hat{\phi}(x) \hat{\phi}^\dagger(y) | \Omega \rangle \\ &= \theta(x^0 - y^0) \langle \Omega | \hat{\phi}(x) \hat{\phi}^\dagger(y) | \Omega \rangle + \theta(y^0 - x^0) \langle \Omega | \hat{\phi}(y)^\dagger \hat{\phi}(x) | \Omega \rangle, \end{aligned} \quad (17.5)$$

$$f(x) = \frac{1}{x \pm i\varepsilon} = \frac{1}{x} \mp i\pi\delta(x),$$

with ε implying the limit to zero. Below, we discuss the

The solution is

$$G(x, y) = \frac{1}{(2\pi)^4} \int d^4p \frac{e^{-ip(x-y)}}{p^2 - m^2 \pm i\varepsilon},$$

Propagator

- Example in L&B how $\theta(t - t') \rightarrow 1/(x + i\epsilon)$

$$\begin{aligned}
 [\Delta(x, y)]_{(2)} &\equiv \theta(y^0 - x^0) \langle 0 | \hat{\phi}^\dagger(y) \hat{\phi}(x) | 0 \rangle \\
 &= \theta(y^0 - x^0) \int \frac{d^3 p}{(2\pi)^3 (2E_p)} e^{-iE_p(y^0 - x^0) + i\mathbf{p} \cdot (\mathbf{y} - \mathbf{x})} \\
 &= i \int_{-\infty}^{\infty} \frac{dz d^3 p}{(2\pi)^4 (2E_p)} \frac{e^{-i(E_p + z)(y^0 - x^0) + i\mathbf{p} \cdot (\mathbf{y} - \mathbf{x})}}{z + i\epsilon}. \quad (17.20)
 \end{aligned}$$

The substitution is now $z' = z + E_p$, giving

$$[\Delta(x, y)]_{(2)} = i \int_{-\infty}^{\infty} \frac{dz' d^3 p}{(2\pi)^4 (2E_p)} \frac{e^{-iz'(y^0 - x^0) + i\mathbf{p} \cdot (\mathbf{y} - \mathbf{x})}}{p^0 - E_p + i\epsilon}. \quad (17.21)$$

Lastly, make the substitution $p \rightarrow -p$, which is to say $(p^0, \mathbf{p}) \rightarrow (-p^0, -\mathbf{p})$, we get

$$\begin{aligned}
 [\Delta(x, y)]_{(2)} &= -i \int_{-\infty}^{\infty} \frac{dz' d^3 p}{(2\pi)^4 (2E_p)} \frac{e^{-ip^0(x^0 - y^0) + i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})}}{p^0 + E_p - i\epsilon} \\
 &= -i \int_{-\infty}^{\infty} \frac{d^4 p}{(2\pi)^4 (2E_p)} \frac{e^{-ip \cdot (x - y)}}{p^0 + E_p - i\epsilon}. \quad (17.22)
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The full propagator $\Delta(x, y) = [\Delta(x, y)]_{(1)} + [\Delta(x, y)]_{(2)}$ is, therefore, the sum of eqns 17.19 and 17.22:

$$\begin{aligned}
 \Delta(x, y) &= i \int \frac{d^4 p}{(2\pi)^4 (2E_p)} e^{-ip \cdot (x - y)} \left(\frac{1}{p^0 - E_p + i\epsilon} - \frac{1}{p^0 + E_p - i\epsilon} \right) \\
 &= \int \frac{d^4 p}{(2\pi)^4} e^{-ip \cdot (x - y)} \frac{i}{(p^0)^2 - E_p^2 + i\epsilon}. \quad (17.23)
 \end{aligned}$$

Breit Wigner

- Many fast decaying particles are resonances.

$$R(\omega) = \frac{\Omega^2 \Gamma^2}{(\omega^2 - \Omega^2)^2 + \Omega^2 \Gamma^2} \quad (9.8)$$

that tells us the general dependence of the average energy of the atom independently of its mass, and the amplitude of the driving force. The response function is a maximum when $\omega = \Omega$, and we can approximate its behavior near the maximum via

$$R(\omega) \simeq \frac{\left(\frac{\Gamma}{2}\right)^2}{(\omega - \Omega)^2 + \left(\frac{\Gamma}{2}\right)^2} \quad (9.9)$$

an expression known as the *Breit-Wigner shape function* [96].

Applying these ideas to particle physics, we can regard the ground state of the atom as one kind of “particle” and an excited level as another kind of “particle”. This may sound a bit strange at first, but each of these states of the atom has their own characteristic energy, frequency and lifetime, where the latter is infinite for the ground state and finite for all of the excited states.

BW $\rightarrow$ Propagator

- The propagator naturally accounts for resonances
 - higher order effects in blob $\rightarrow$ finite width added to $i\epsilon$
 - RADIATIVE EFFECTS

$\hat{\phi}^\dagger(y)$ creates precisely one quantum from the vacuum $|0\rangle$ (and later we destroy it). When we turn on the interactions it turns out¹² that the full propagator resembles the free version. The general form of the two-point propagator in momentum space is

$$\tilde{G}^{(2)}(p) = \frac{iZ_{\mathbf{p}}}{p^2 - m^2 - \tilde{\Sigma}(p) + i\Gamma_{\mathbf{p}}} + \left(\begin{array}{c} \text{Multiparticle} \\ \text{parts} \end{array} \right), \quad (17.35)$$

where

- $Z_{\mathbf{p}}$ tells us how likely it is for a particle with momentum $\mathbf{p}$ to exist stably without being destroyed by the interactions,
- $\tilde{\Sigma}(p)$ tells us about how the particle of momentum p interacts with the vacuum,
- $(2\Gamma_{\mathbf{p}})^{-1}$ is the particle's lifetime,
- The full propagator also contains a contribution from short-lived multiparticle states which can be scattered out of the vacuum.

Thomson's Take

The Breit-Wigner Resonance

- ★ Need to consider carefully the propagator term $1/(s - m_Z^2)$ which diverges when the C.o.M. energy is equal to the rest mass of the Z boson
- ★ To do this need to account for the fact that the Z boson is an unstable particle
 - For a stable particle at rest the time development of the wave-function is:

$$\psi \sim e^{-imt}$$

- For an unstable particle this must be modified to

$$\psi \sim e^{-imt} e^{-\Gamma t/2}$$

so that the particle probability decays away exponentially

$$\psi^* \psi \sim e^{-\Gamma t} = e^{-t/\tau} \quad \text{with} \quad \tau = \frac{1}{\Gamma_Z}$$

- Equivalent to making the replacement

$$m \rightarrow m - i\Gamma/2$$

- ★ In the Z boson propagator make the substitution:

$$m_Z \rightarrow m_Z - i\Gamma_Z/2$$

- ★ Which gives:

$$(s - m_Z^2) \longrightarrow [s - (m_Z - i\Gamma_Z/2)] = s - m_Z^2 + im_Z\Gamma_Z + \frac{1}{4}\Gamma_Z^2 \approx s - m_Z^2 + im_Z\Gamma_Z$$

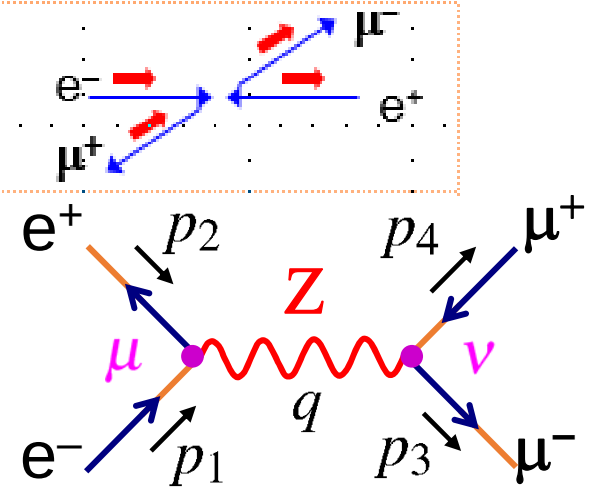
where it has been assumed that $\Gamma_Z \ll m_Z$

- ★ Which gives

$$\left| \frac{1}{s - m_Z^2} \right|^2 \rightarrow \left| \frac{1}{s - m_Z^2 + im_Z\Gamma_Z} \right|^2 = \frac{1}{(s - m_Z^2)^2 + m_Z^2\Gamma_Z^2}$$

Example Z resonance Lifetime 10^{23-2}

$$M_{RR} = -\frac{g_Z^2}{q^2 - m_Z^2} c_R^e c_R^\mu g_{\mu\nu} [\bar{v}_\downarrow(p_2) \gamma^\mu u_\uparrow(p_1)] [\bar{u}_\uparrow(p_3) \gamma^\nu v_\downarrow(p_4)]$$



Remember: the L/R refer to the helicities of the initial/final state particles

★ Fortunately we have calculated these terms before when considering $e^+e^- \rightarrow \gamma \rightarrow \mu^+\mu^-$ giving: (pages 137-138)
 $[\bar{v}_\downarrow(p_2) \gamma^\mu u_\uparrow(p_1)] [\bar{u}_\uparrow(p_3) \gamma^\nu v_\downarrow(p_4)] \doteq s(1 + \cos \theta)$ etc.

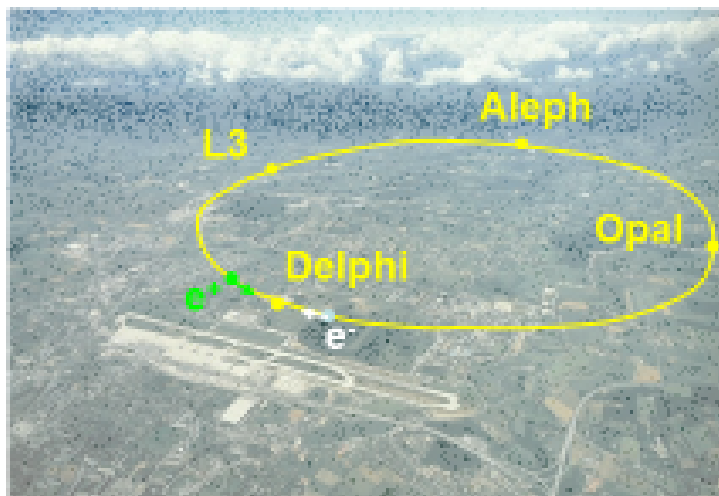
“Radiative effects”...

★ And the Matrix elements become

$$|M_{RR}|^2 = \frac{g_Z^4 s^2}{(s - m_Z^2)^2 + m_Z^2 \Gamma_Z^2} (c_R^e)^2 (c_R^\mu)^2 (1 + \cos \theta)^2 \quad \text{etc.}$$

Electroweak Measurements at LEP

- ★ The **L**arge **E**lectron **P**ositron (LEP) Collider at CERN (1989-2000) was designed to make precise measurements of the properties of the Z and W bosons.



- 26 km circumference accelerator straddling French/Swiss border
- Electrons and positrons collided at 4 interaction points
- 4 large detector collaborations (each with 300-400 physicists):
ALEPH,
DELPHI,
L3,
OPAL

Basically a large Z and W factory:

- ★ 1989-1995: **Electron-Positron collisions at $\sqrt{s} = 91.2$ GeV**
 - 17 Million **Z bosons** detected
- ★ 1996-2000: **Electron-Positron collisions at $\sqrt{s} = 161$ -208 GeV**
 - 30000 **W⁺W⁻ events** detected

Photon $\rightarrow$ Resonance Z

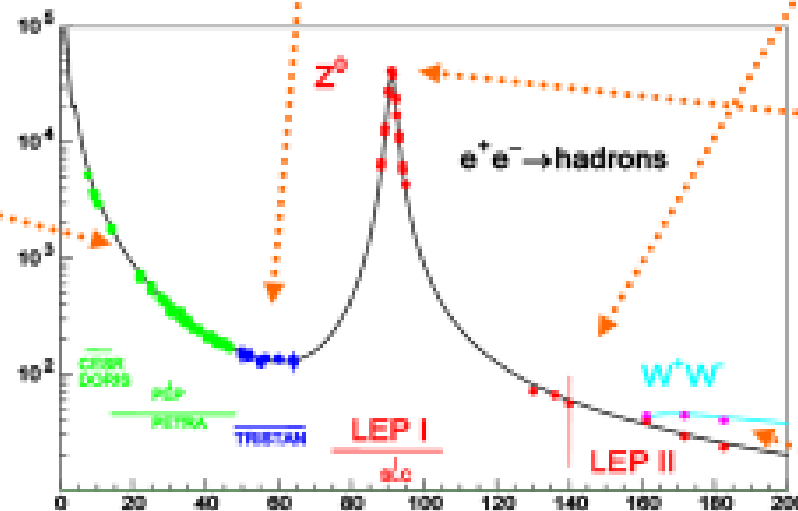
e^+e^- Annihilation in Feynman Diagrams

In general e^+e^- annihilation involves both photon and Z exchange : + interference

$$\left| \begin{array}{c} e^+ \\ e^- \end{array} \right\rangle \gamma \left| \begin{array}{c} \bar{f} \\ f \end{array} \right\rangle + \left| \begin{array}{c} e^+ \\ e^- \end{array} \right\rangle Z \left| \begin{array}{c} \bar{f} \\ f \end{array} \right\rangle \right|^2$$

$$\left| \begin{array}{c} e^+ \\ e^- \end{array} \right\rangle \gamma \left| \begin{array}{c} \bar{f} \\ f \end{array} \right\rangle \right|^2$$

Well below Z: photon exchange dominant



$$\left| \begin{array}{c} e^+ \\ e^- \end{array} \right\rangle Z \left| \begin{array}{c} \bar{f} \\ f \end{array} \right\rangle \right|^2$$

At Z resonance: Z exchange dominant

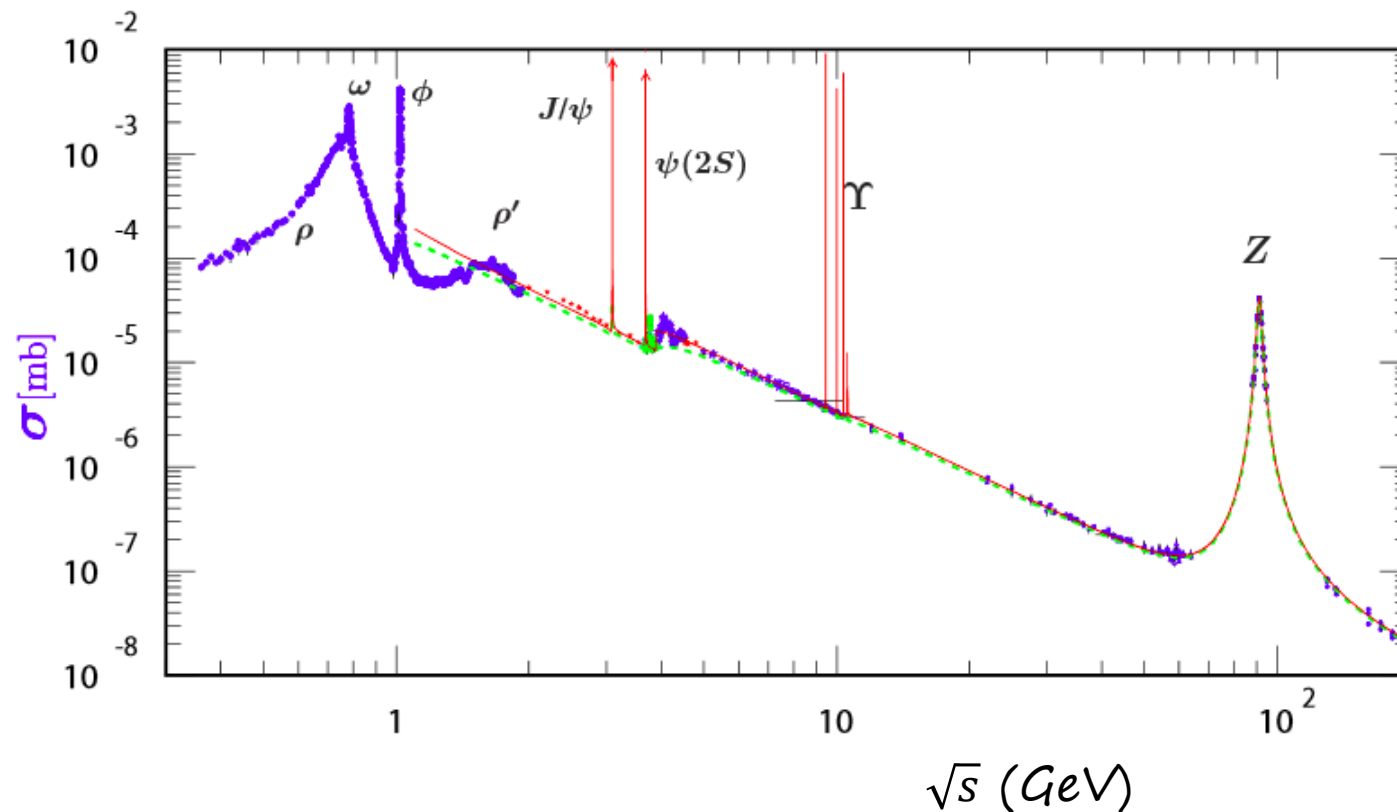
High energies: WW production

$$\left| \begin{array}{c} e^+ \\ e^- \end{array} \right\rangle \gamma \begin{array}{c} W^+ \\ W^- \end{array} + \left| \begin{array}{c} e^+ \\ e^- \end{array} \right\rangle Z \begin{array}{c} W^+ \\ W^- \end{array} + \left| \begin{array}{c} e^+ \\ e^- \end{array} \right\rangle \nu_e \begin{array}{c} W^+ \\ W^- \end{array} \right|^2$$

Resonances!

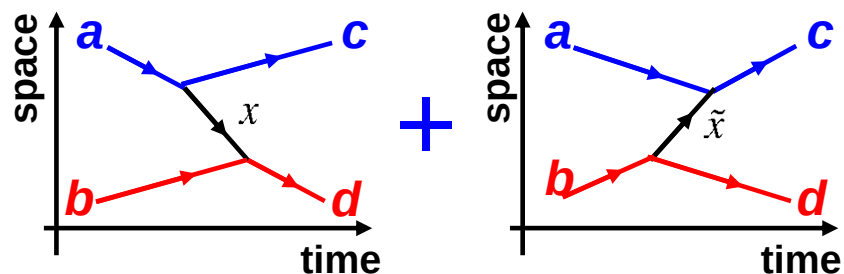
- Little treatment of these in particle physics books
 - But super prevalent part of particle physics!
- Take a look at web browser for other resonances

σ and R in e^+e^- Collisions



Virtual Particles

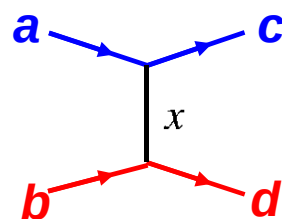
“Time-ordered QM”



- Momentum conserved at vertices
- Energy **not** conserved at vertices
- Exchanged particle **“on mass shell”**

$$E_x^2 - |\vec{p}_x|^2 = m_x^2$$

Feynman diagram



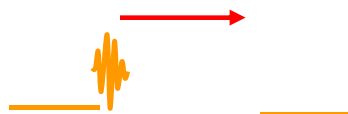
$$M_{fi} = \frac{g_a g_b}{q^2 - m_x^2}$$

- Momentum **AND** energy conserved at interaction vertices
- Exchanged particle **“off mass shell”**

$$E_x^2 - |\vec{p}_x|^2 = q^2 \neq m_x^2$$

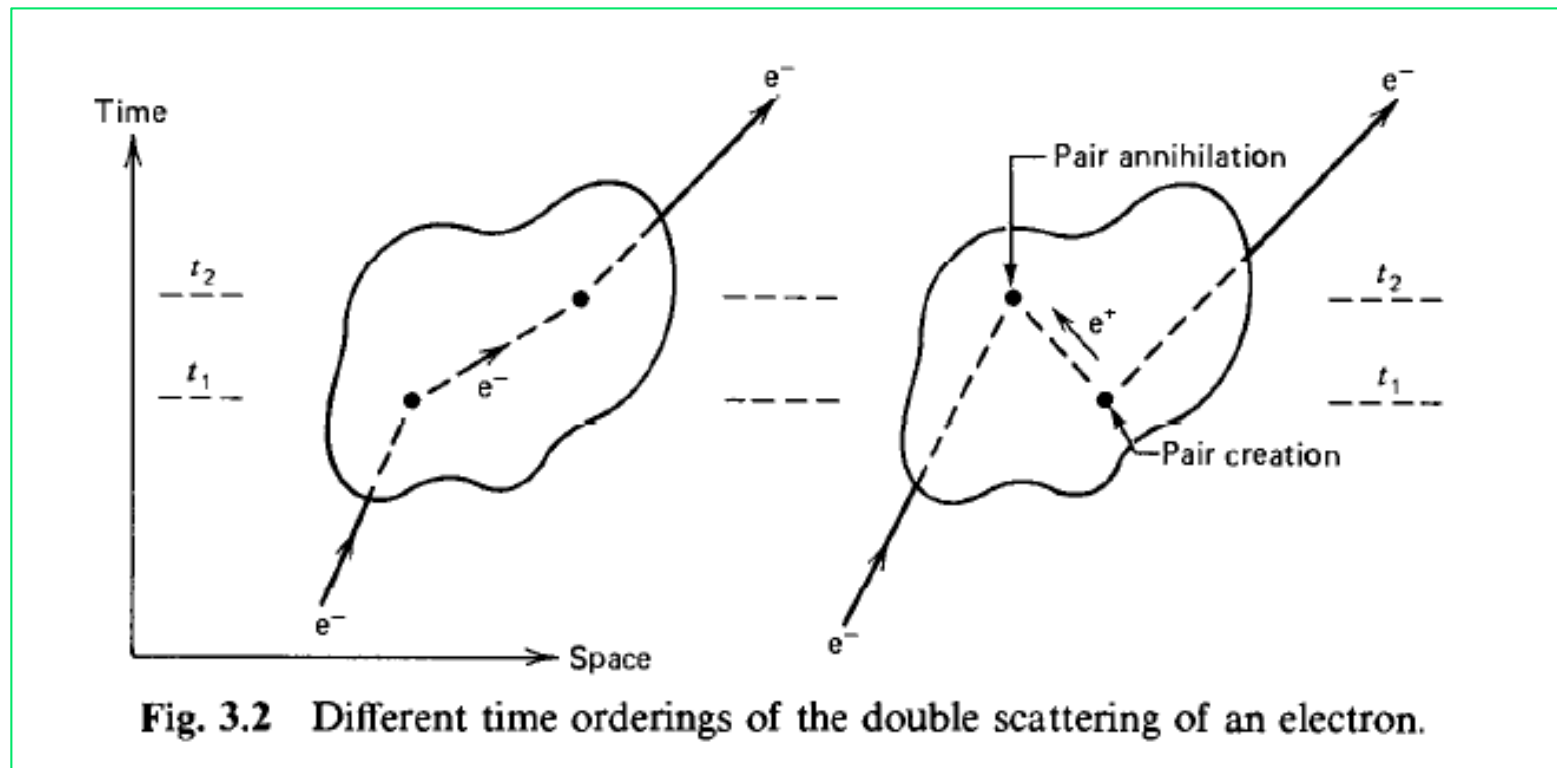
VIRTUAL PARTICLE

- Can think of observable “on mass shell” particles as propagating waves and unobservable virtual particles as normal modes between the source particles:



Antiparticles Bkwd in Time

- Cool from Halzen and Martin
- Explains why “backwards in time” solutions make sense



- left and right in diagram are indistinguishable $\rightarrow$ 2nd case is same as electron traveling backwards in time (variations on that idea)
 - Maybe not strictly same as Feynmans(?), but at least cool mnemonic

Example Toy (ABC) Theory Feynman Rules

• Feynman Rules to Calculate M

Our problem is to find the amplitude $\mathcal{M}$ associated with a given Feynman diagram. The ritual is as follows [5]:

1. **Notation:** Label the incoming and outgoing four-momenta $p_1, p_2, \dots, p_n$ (Figure 6.6). Label the internal momenta $q_1, q_2, \dots$. Put an arrow beside each line, to keep track of the 'positive' direction (forward in time for external lines, arbitrary for internal lines).
2. **Vertex factors:** For each vertex, write down a factor

$$-ig$$

g is called the *coupling constant*; it specifies the strength of the interaction between A, B, and C. In this toy theory, g has the dimensions of momentum; in the 'real-world' theories, we shall encounter later on, the coupling constant is always dimensionless.

3. **Propagators:** For each internal line, write a factor

$$\frac{i}{q_j^2 - m_j^2 c^2}$$

where q_j is the four-momentum of the line and m_j is the mass of the particle the line describes. (Note that $q_j^2 \neq m_j^2 c^2$, because a virtual particle does not lie on its mass shell.)

4. **Conservation of energy and momentum:** For each vertex, write a delta function of the form

$$(2\pi)^4 \delta^4(k_1 + k_2 + k_3)$$

where the k 's are the three four-momenta coming into the vertex (if the arrow leads outward, then k is minus the four-momentum of that line). This factor imposes conservation of energy and momentum at each vertex since the delta function is zero unless the sum of the incoming momenta equals the sum of the outgoing momenta.

5. **Integration over internal momenta:** For each internal line, write down a factor^a

$$\frac{1}{(2\pi)^4} d^4 q_j$$

and integrate over all internal momenta.

6. **Cancel the delta function:** The result will include a delta function

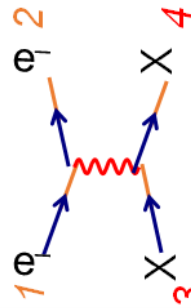
$$(2\pi)^4 \delta^4(p_1 + p_2 + \dots - p_n)$$

reflecting overall conservation of energy and momentum. Erase this factor^b and multiply by i . The result is $\mathcal{M}$.

Summarized

- 1) Diagram – Notate (1,2,3,etc)
- 2) Each Vertex factor F – coupling g
 $\rightarrow F = -ig$
- 3) (Each) propagator $i/(q^2 - m^2)$
- 4) Each vertex ("uvw") $\delta(p_u \pm p_v \pm p_w)$ ie 4v mom conservation 3legs/arrows
- 5) Integrate internal lines $d^4 q$
- 6) Cancel/erase the final delta fn (δ^4)
 --global 4v mom consv
- 7) Multiply by i

tree diagram δ fn's: 2 (8) δ fns [two 4v] – 1(4)
 Integrals – 1 (4) final = no "real" integrations!



Decay Example - trivial

- HOAKY?

< Back

Q A



Introduction to Elementary Particles

David Griffiths

Expand all | Collapse all

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6.2 The Golden Rule 203

6.3 Feynman Rules for a Toy Theory 211

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6.3.1 Lifetime of the A

The simplest possible diagram, representing the lowest-order contribution to $A \rightarrow B + C$, has no internal lines at all ([Figure 6.7](#)). There is one vertex, at which we pick up a factor of $-ig$ (Rule 2) and a delta function

$$(2\pi)^4 \delta^4(p_1 - p_2 - p_3)$$

(Rule 4), which we promptly discard (Rule 6). Multiplying by i , we get

$$M = g \quad (6.50)$$

This is the *amplitude* (to lowest order); the decay rate is found by plugging M into [Equation 6.35](#):

$$\Gamma = \frac{g^2 |\mathbf{p}|}{8\pi \hbar m_A^2 c} \quad (6.51)$$

where $|\mathbf{p}|$ (the magnitude of either outgoing momentum) is

$$|\mathbf{p}| = \frac{c}{2m_A} \sqrt{m_A^4 + m_B^4 + m_C^4 - 2m_A^2 m_B^2 - 2m_A^2 m_C^2 - 2m_B^2 m_C^2} \quad (6.52)$$

The lifetime of the A , then, is

$$\tau = \frac{1}{\Gamma} = \frac{8\pi \hbar m_A^2 c}{g^2 |\mathbf{p}|} \quad (6.53)$$

You should check for yourself that τ comes out with the correct units.

Mann → Symmetric B+B

- Decay

where $\vec{p}$ is the magnitude of the outgoing momenta of the final particles and $S = \frac{1}{2!}$ since the final particles are identical. Conservation due to the δ -function implies

$$m_A = \sqrt{|\vec{p}|^2 + m_B^2} + \sqrt{|\vec{p}|^2 + m_B^2}$$

$$\Rightarrow p = \sqrt{\frac{m_A^2}{4} - m_B^2} \quad (10.6)$$

and so we get

$$\Gamma = \frac{g^2}{32\pi\hbar cm_A} \sqrt{1 - 4\frac{m_B^2}{m_A^2}} \quad (10.7)$$

for the decay rate of the A particle.

For any given theory the masses m_A and m_B are fixed, reflecting the fact that in the real world an elementary particle cannot change its rest mass. However, we can mathematically adjust the mass to be whatever we like – in so doing we are really comparing a whole continuum of theories dependent on the choice of the particle masses. We can also do the same for the couplings. Such comparisons afford important physical insight.

For the decay process above, we see that the smaller the value of g , the smaller the decay rate, in accord with our intuition that a weaker coupling makes for a less probable interaction or a less likely decay. If $m_B > \frac{m_A}{2}$ (i.e. if the B particle has a rest mass more than half that of the A), then the decay

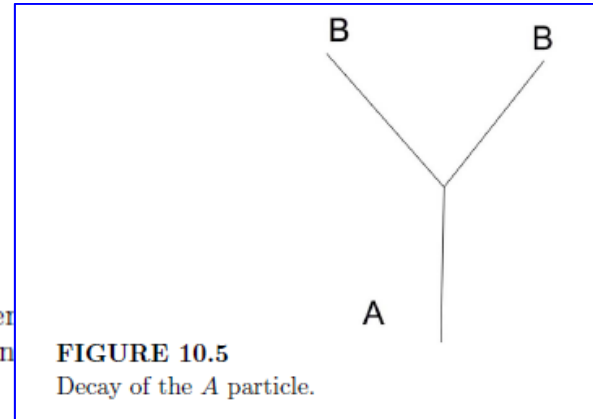


FIGURE 10.5
Decay of the A particle.

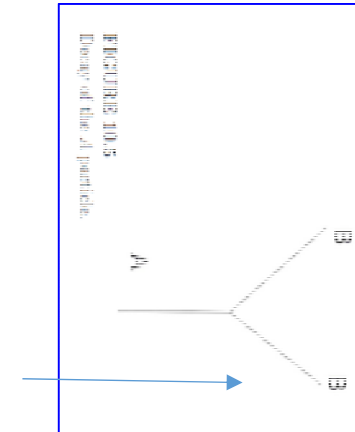
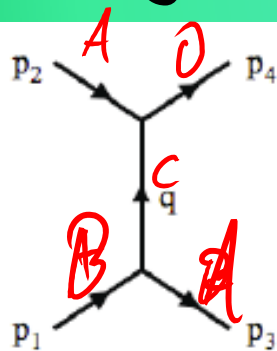
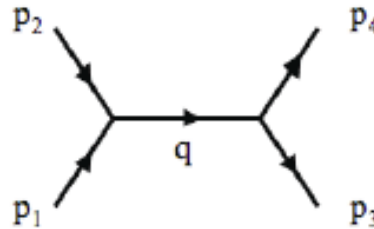


FIGURE 10.5
Decay of the A particle.

Toy Theory $A+B \rightarrow A+B$



M_1



M_2

$$M_1 \rightarrow \int (-ig)^2 \left(\frac{i}{q^2 - m_c^2 c^2} \right) (2\pi)^4 \delta^4(p_1 - q - p_3) (2\pi)^4 \delta^4(p_2 + q - p_4) \frac{d^4 q}{(2\pi)^4}$$

do integral $\rightarrow$ apply 2nd δ fn to first $\rightarrow$ sends $q = p_4 - p_2$

$$\begin{aligned} \delta &\rightarrow \delta^4(p_1 - (p_4 - p_2) - p_3) \\ &= \delta^4(p_1 + p_2 - p_3 - p_4) \end{aligned}$$

Continuing.. M_1 $A+B \rightarrow A+B$

$$\frac{-ig^2}{(p_1 - p_3)^2 - m_c^2 c^2} (2\pi)^4 \delta^4(p_1 + p_2 - p_3 - p_4) = -i\mathcal{M}_1$$

Very common way to denote rule 7 – many text derive M starting with this expression due to other reasons

$$\mathcal{M}_1 = \frac{g^2}{(p_1 - p_3)^2 - m_c^2 c^2}$$

Done with M_1

M_2 for $A+B \rightarrow A+B$

- (Common move also:) BUT we don't need to do everything again.
- Almost nothing would be different except delta fn's and what q is

$$M_2 \rightarrow \int (-ig)^2 \left(\frac{i}{q^2 - m_c^2 c^2} \right) (2\pi)^4 \delta^4(p_1 - q - p_3) (2\pi)^4 \delta^4(p_2 + q - p_4) \frac{d^4 q}{(2\pi)^4}$$

$\delta(p_1 + p_2 - q) \quad \delta(q - p_3 - p_4)$

$q = (p_1 + p_2)$

$$\mathcal{M}_2 = \frac{g^2}{(p_1 + p_2)^2 - m_c^2 c^2}$$

Squaring

- Total M has two terms, there is an interference term when modulus taken
- Note these M 's are real (not complex)...this will not always be the case

$$|M|^2 = (M_1 + M_2)(M_1^* + M_2^*) = |M_1|^2 + |M_2|^2 + \underbrace{2\text{Re}[M_1 M_2^*]}_{\text{Interference}}$$

$$\frac{M}{g^2} = \frac{1}{t - m_c^2} + \frac{1}{s - m_c^2}$$

- For $m = 0$ (like a photon)

$$|M|^2 = \frac{g^2}{t^2} + \frac{g^2}{s^2} + \frac{2g^2}{ts}$$

Invariant! (by construction)

Putting Into Cross Section

- Which x.s formula? Well we must choose frame... however our for ~all formula still need COM momentum calc'd (used in all expressions :)

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar c}{8\pi}\right)^2 \frac{S|\mathcal{M}|^2}{(E_1 + E_2)^2} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}$$

$(\sqrt{s})^2$

$$\sigma = \frac{1}{64\pi^2 s} \frac{|\vec{p}_f^*|}{|\vec{p}_i^*|} \int |M_{fi}|^2 d\Omega^*$$

$$\frac{d\sigma}{dt} = \frac{1}{64\pi s |\vec{p}_i^*|^2} |M_{fi}|^2$$

In COM frame AND with $m_A = m_B$ (like last class)

$$p_i = \sqrt{\frac{s}{4} - m_B^2}$$

$$\frac{d\sigma}{dt} = \frac{g^2}{32\pi s \sqrt{s - 4m^2}} \left(\frac{1}{t^2} + \frac{1}{s^2} + \frac{2}{st} \right)$$

- For $d\Omega$ formulae $p_f = p_i$ for $m_A = m_B$

$$\frac{d\sigma}{d\Omega^*} = \frac{1}{64\pi^2 s} \left(\frac{1}{t^2} + \frac{1}{s^2} + \frac{2}{st} \right)$$

Note that t in this case ($m_A = m_B$) in previous pset shown to be $t \propto 1 - \cos\theta \rightarrow$ x.s. Diverges at $\theta = 0$!!
Common $\rightarrow$ QED/QCD ($m_{\text{prop}} = 0$) x.s. generally strongly forward peaked

$A+A \rightarrow B+B$

- See Griff for the other main example
- Interesting to point out some features on that one because of crossing

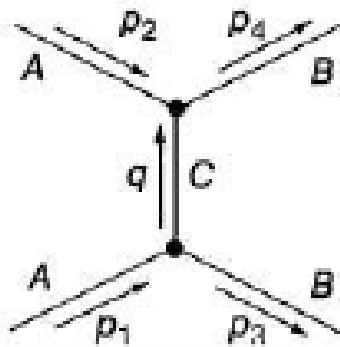


Fig. 6.8 Lowest-order contribution

to $A + A \rightarrow B + B$.

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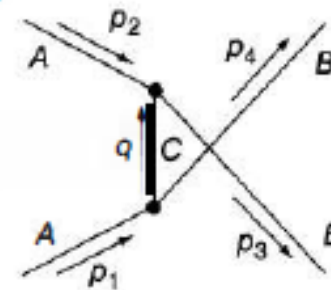
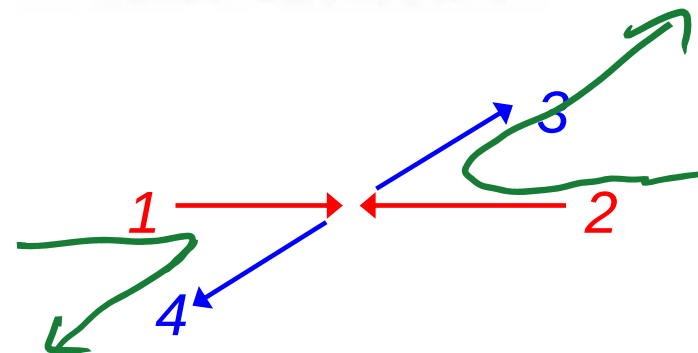
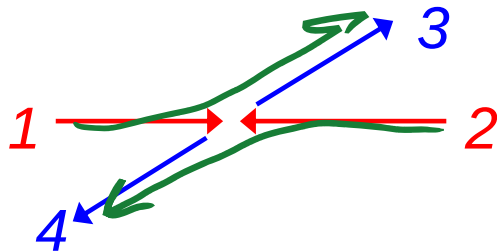
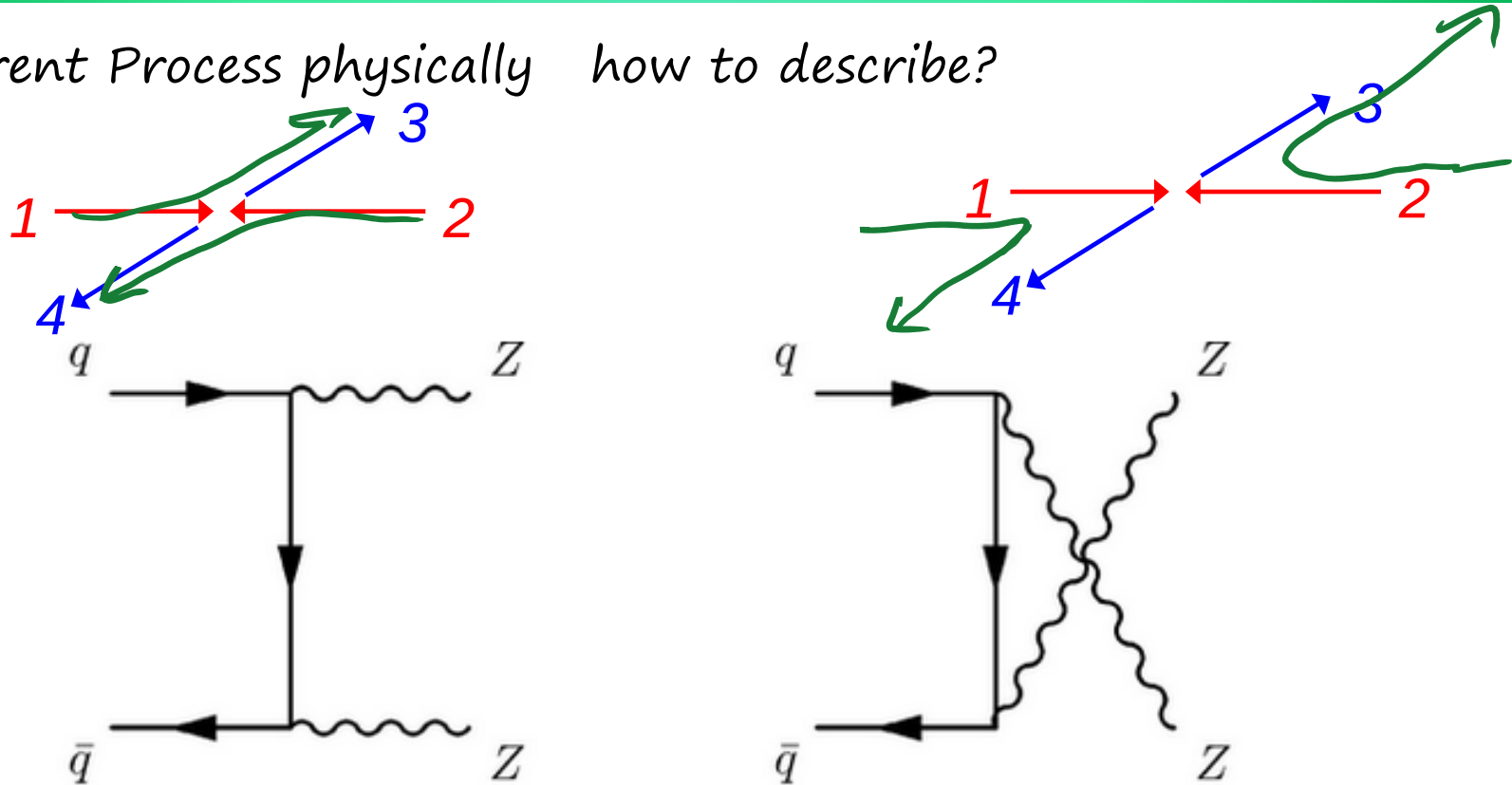


Fig. 6.9 Second diagram contributing in lowest order to $A + A \rightarrow B + B$.



REVIEW Crossed Diagrams

- Different Process physically how to describe?

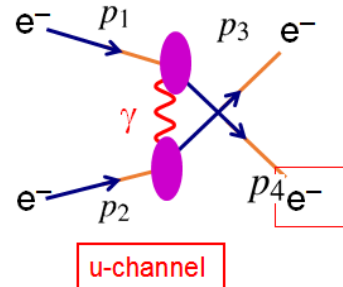
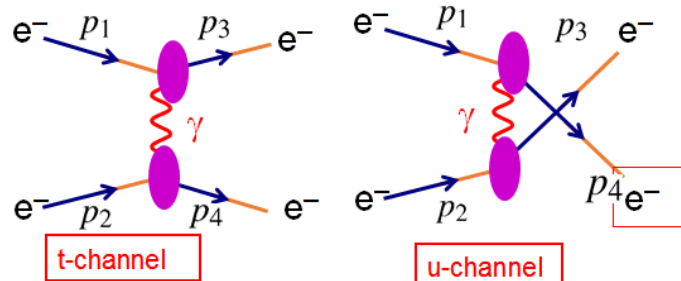
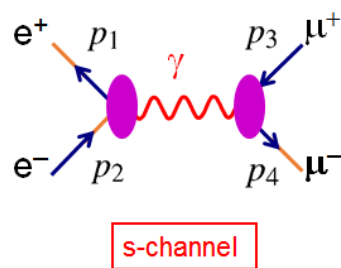
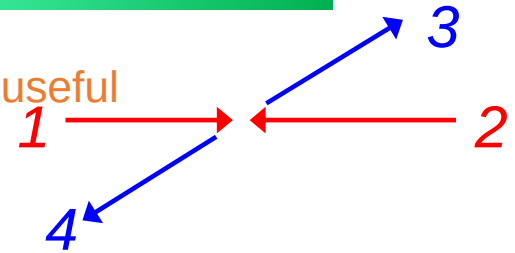


- The top assignments aren't unambiguously associated with t vs u , Left vs Right (L/R), but...
- I THINK: it only makes sense to consider u (when possible to cross) in comparison to " t ", so...
- So think of t and u together, for every contribution for t channel on left, there is a contribution from u like the one on the right.
 - Compared to when t/u have the opposite assignment L vs R on top, the L- u vs the L- t (e.g.) are closely related, but different by a simple thing e.g. phase factor or $\pm$ trig function, etc..

Review/New : Mandelstam s, t and u

- ★ In particle scattering/annihilation there are three particularly useful Lorentz Invariant quantities: s, t and u

- ★ Consider the scattering process $1 + 2 \rightarrow 3 + 4$



$$s = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_1 - p_4)^2$$

Note also

$$s = (p_3 + p_4)^2 \quad t = (p_2 - p_4)^2 \quad u = (p_2 - p_3)^2$$

“mom lost = gained”

A few more relations are on the way (besides $s+t+u = \sum m_i^2$)

$A+A \rightarrow B+B$ - Crossing diagrams

- See Griff for this other main example $AA \rightarrow BB$
- Interesting to point out some features on that one because of crossing

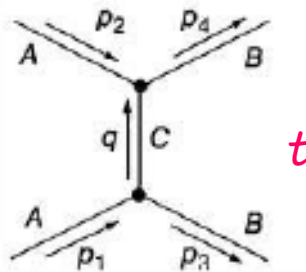


Fig. 6.8 Lowest-order contribution
1 to $A + A \rightarrow B + B$.

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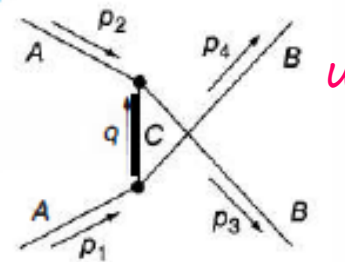
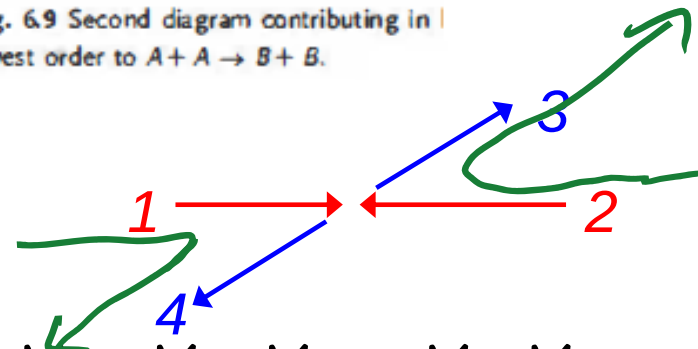
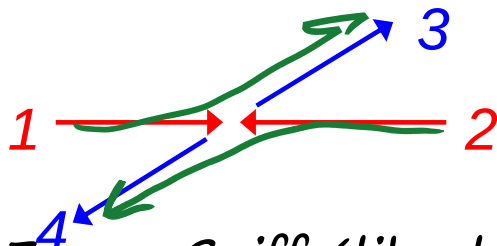


Fig. 6.9 Second diagram contributing in
lowest order to $A + A \rightarrow B + B$.



- From Griff (like last example): $M = M_1 + M_2 = M_t + M_u$
(also no “real” integration – only δ ’s)

$$\mathcal{M} = \frac{g^2}{(p_4 - p_2)^2 - m_c^2 c^2} + \frac{g^2}{(p_3 - p_2)^2 - m_c^2 c^2} = g^2 * \frac{1}{t - m^2} + \frac{1}{u - m_c^2}$$

Example Toy (ABC) Theory Feynman Rules

• Feynman Rules to Calculate M

Our problem is to find the amplitude $\mathcal{M}$ associated with a given Feynman diagram. The ritual is as follows [5]:

1. **Notation:** Label the incoming and outgoing four-momenta $p_1, p_2, \dots, p_n$ (Figure 6.6). Label the internal momenta $q_1, q_2, \dots$. Put an arrow beside each line, to keep track of the 'positive' direction (forward in time for external lines, arbitrary for internal lines).
2. **Vertex factors:** For each vertex, write down a factor

$$-ig$$

g is called the *coupling constant*; it specifies the strength of the interaction between A, B, and C. In this toy theory, g has the dimensions of momentum; in the 'real-world' theories, we shall encounter later on, the coupling constant is always dimensionless.

3. **Propagators:** For each internal line, write a factor

$$\frac{i}{q_j^2 - m_j^2 c^2}$$

where q_j is the four-momentum of the line and m_j is the mass of the particle the line describes. (Note that $q_j^2 \neq m_j^2 c^2$, because a virtual particle does not lie on its mass shell.)

4. **Conservation of energy and momentum:** For each vertex, write a delta function of the form

$$(2\pi)^4 \delta^4(k_1 + k_2 + k_3)$$

where the k 's are the three four-momenta coming into the vertex (if the arrow leads outward, then k is minus the four-momentum of that line). This factor imposes conservation of energy and momentum at each vertex since the delta function is zero unless the sum of the incoming momenta equals the sum of the outgoing momenta.

5. **Integration over internal momenta:** For each internal line, write down a factor^a

$$\frac{1}{(2\pi)^4} d^4 q_j$$

and integrate over all internal momenta.

6. **Cancel the delta function:** The result will include a delta function

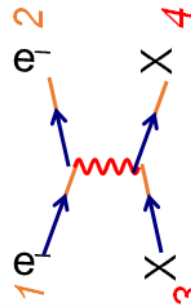
$$(2\pi)^4 \delta^4(p_1 + p_2 + \dots + p_n)$$

reflecting overall conservation of energy and momentum. Erase this factor^b and multiply by i . The result is $\mathcal{M}$.

Summarized

- 1) Diagram – Notate (1,2,3,etc)
- 2) Each Vertex factor F – coupling g
 $\rightarrow F = -ig$
- 3) (Each) propagator $i/(q^2 - m^2)$
- 4) Each vertex ("uvw") $\delta(p_u \pm p_v \pm p_w)$ ie 4v mom conservation 3legs/arrows
- 5) Integrate internal lines $d^4 q$
- 6) Cancel/erase the final delta fn (δ^4)
 --global 4v mom consv
- 7) Multiply by i

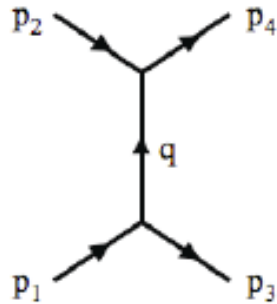
tree diagram δ fn's: 2 (8) δ fns [two 4v] – 1(4)
 Integrals – 1 (4) final = no "real" integrations!



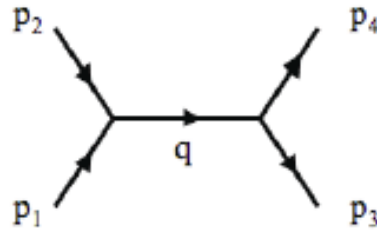
Griffiths – Toy Theory

- Griffiths Uses the Rules to Derive $A+A \rightarrow B+B$ at Tree Level
- I will do the same for $A+B \rightarrow A+B$
- Griffiths also then shows how higher order corrections behave
 - I have some comments about u, t, s and crossing about this case.
- I will only summarize the latter

Toy Theory $A+B \rightarrow A+B$



M_1



M_2

$$M_1 \rightarrow \int (-ig)^2 \left(\frac{i}{q^2 - m_c^2 c^2} \right) (2\pi)^4 \delta^4(p_1 - q - p_3) (2\pi)^4 \delta^4(p_2 + q - p_4) \frac{d^4 q}{(2\pi)^4}$$

do integral $\rightarrow$ apply 2nd δ fn to first $\rightarrow$ sends $q = p_4 - p_2$

$$\begin{aligned} \delta &\rightarrow \delta^4(p_1 - (p_4 - p_2) - p_3) \\ &= \delta^4(p_1 + p_2 - p_3 - p_4) \end{aligned}$$

Continuing.. M_1 $A+B \rightarrow A+B$

$$\frac{-ig^2}{(p_1 - p_3)^2 - m_c^2 c^2} (2\pi)^4 \delta^4(p_1 + p_2 - p_3 - p_4) = -i\mathcal{M}_1$$

Very common way to denote rule 7 – many text derive M starting with this expression due to other reasons

$$\mathcal{M}_1 = \frac{g^2}{(p_1 - p_3)^2 - m_c^2 c^2}$$

Done with M_1

crossing t/u

- Now we see a concrete case

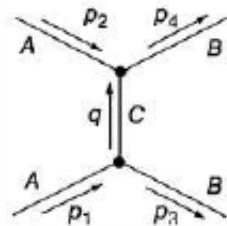
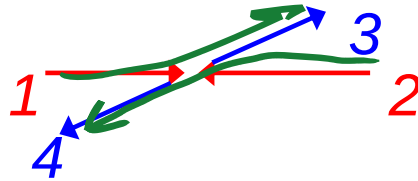


Fig. 6.8 Lowest-order contribution to $A + A \rightarrow B + B$.

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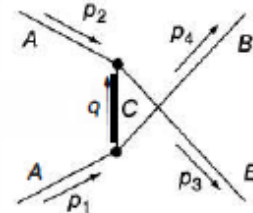
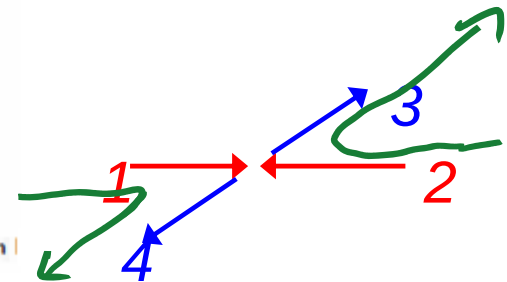


Fig. 6.9 Second diagram contributing in lowest order to $A + A \rightarrow B + B$.



$$\mathcal{M} = \frac{g^2}{(p_4 - p_2)^2 - m_c^2 c^2} + \frac{g^2}{(p_3 - p_2)^2 - m_c^2 c^2}$$

$$= g^2 * \frac{1}{t - m^2} + \frac{1}{u - m_c^2}$$

- Choose again $m_A = m_B$; $m_C = 0$ as in Griff ($\propto 1/t + 1/u$)
- Recall Pset again $t/u = 2p(1 \pm \cos\theta) \rightarrow$

- Point 2) : for every angle there are both t and u contributing (QM'ally cannot distinguish) then there is even an **interference**
 - $M^2 \propto 1/u^2 + 1/t^2 + \underline{2/tu}$ see 2 slides from now, e.g. $\theta=0$, $1/t$ diverges, $1/u$ finite at $\theta = 180$ $1/u$ diverges $1/t$ finite

Final calc

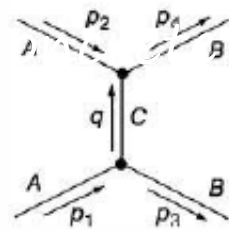
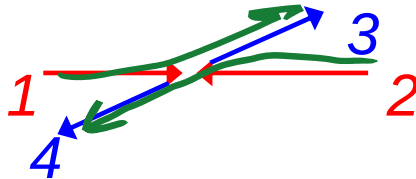


Fig. 6.8 Lowest-order contribution to $A + A \rightarrow B + B$.

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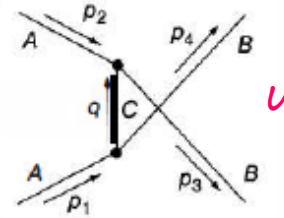
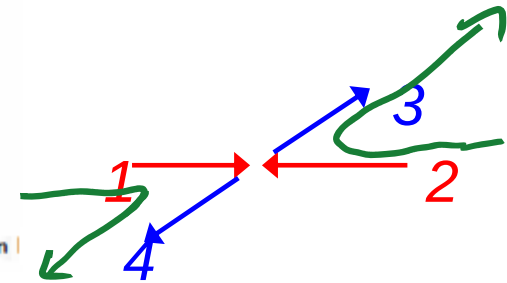


Fig. 6.9 Second diagram contributing in lowest order to $A + A \rightarrow B + B$.



$$\mathcal{M} = \frac{g^2}{(p_4 - p_2)^2 - m_c^2 c^2} + \frac{g^2}{(p_3 - p_2)^2 - m_c^2 c^2}$$

$$= g^2 * \frac{1}{t - m^2} + \frac{1}{u - m_c^2}$$

- Comment on t/u coming.... first calc
- Choose again $m_A = m_B$; $m_c = 0$ as in Griff (now $M \propto 1/t + 1/u$)
- Recall Pset again $t/u = 2p(1 \pm \cos\theta) \rightarrow$

$$\mathcal{M} = -\frac{g^2}{p^2 \sin^2 \theta} \quad \frac{d\sigma}{d\Omega} = \frac{1}{2} \left(\frac{\hbar c g^2}{16\pi E p^2 \sin^2 \theta} \right)^2$$

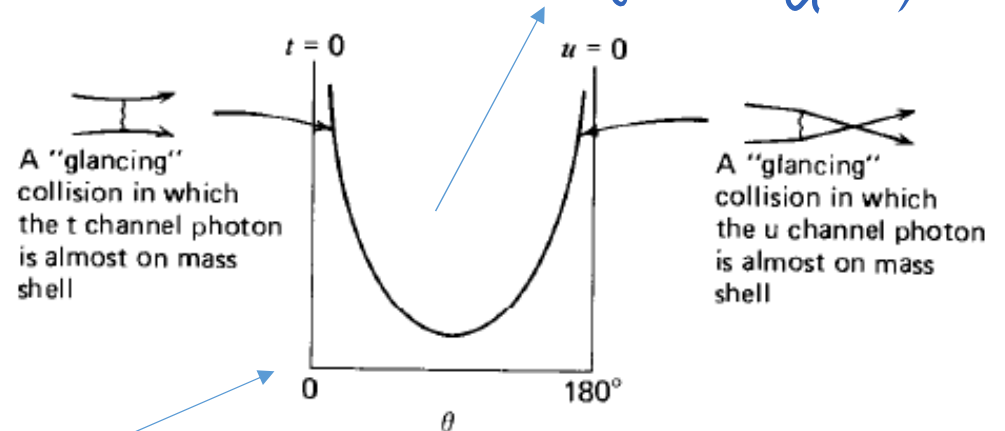
- Very Similar form as Rutherford Scattering (Grad Lab)
- OBVIOUSLY Mandelstam forms are convenient but for usefulness they generally need converted into ANGLE formula
 - Having a proficiency in both is good (angle $\leftrightarrow$ s, t, u)

Infrared Divergences/Safety

- **Point 3)** Note that the infinity may not be “physical” – Griff’s comments earlier (EM infinite range)
 - There must be some limit, practically
- **Divergence** not ~same as that which renormalization fixes!
 - (those are “UV” diverge as $q \rightarrow \infty$; these: as $q \rightarrow 0$)
- Theoretically inconvenient $\rightarrow$ “Regulate” similar to renormalization
 - $\rightarrow$ tools/methods to render these expressions **Infrared “SAFE”**
 - e.g. small mass -- More on this later probably – probably not best example

$$\mathcal{M} = -\frac{g^2}{p^2 \sin^2 \theta} \quad \frac{d\sigma}{d\Omega} = \frac{1}{2} \left(\frac{\hbar c g^2}{16\pi E p^2 \sin^2 \theta} \right)^2 \propto \left(\frac{1}{t} + \frac{1}{u} \right)$$

$$\frac{d\sigma}{d\Omega}^* = \frac{1}{64\pi^2 s} \left(\frac{1}{t^2} + \frac{1}{s^2} + \frac{2}{st} \right)$$

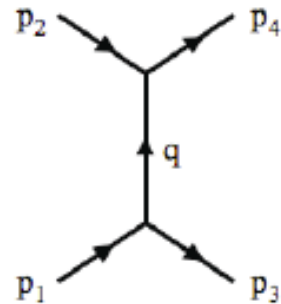


This side is
forward
direction

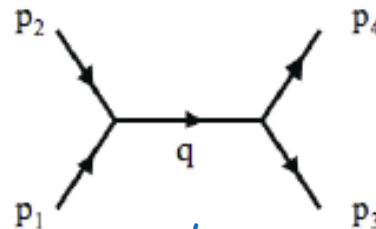
Fig. 4.8 The differential cross section, $d\sigma/d\Omega$, for electron-electron scattering.

Halzen and Martin

Crossing/Interference/etc **NOTE:** $A+B \rightarrow A+B$



M_1



not
crossing

M_2

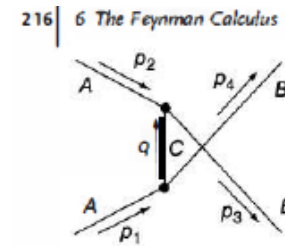


Fig. 6.9 Second diagram contributing in lowest order to $A+A \rightarrow B+B$.

- Note that our $AB \rightarrow AB$ case was **not** a crossing example – rather interference
 - can't have crossing because particles aren't identical
- Interference $\neq$ crossing -- true however: if cross /u there will always be interference (pretty sure)
- u channel will not exist unless t-channel does
 - But it won't always (e.g. non-identical particles) when t-channel exists
- Some cases where t and s-channels can interfere
- Can have more than 2 diagrams in M then multiple interference
 - Although frankly never seen it and I don't think there are any examples with $n > 2$ in Griff
- Crossing requires identical particles (to be crossed)
 - So indeed it is at least a tip off that your S statistical factor isn't just 1

Note: Feynman Rules-Shmules?

- We've already done the δ fn counting for tree level (ie $2 \rightarrow 2$ no loops) diagrams btw $2 \rightarrow 3$ is NLO $\rightarrow$ UV divergence
 - No real integral methods needed!!! NLO/loops are beyond the scope of our course (ability to calc/renorm not too useful e.g. to practical experimentalists—(maybe a few exceptions like $2 \rightarrow 3$) –but generally a very theorist-specialized skill)
- The complex expressions used for M with all the rules, especially with the INTEGRALS are really only needed for $\geq$ NLO/loops
- Many other courses (e.g. Halzen and Martin) start with simpler motivation for tree level calcs from Perturb. Theory (just as I have reminded you of last time) in a way that doesn't involve ANY INTEGRATION !!!
- This is in the context of “**SPINLESS**” QED which is a very nice thing to study as well – so we will, a little too
 - Good intuition before Dirac complications for many phenomena
 - Not

Matrix Element again – focus

- Matrix element so much easier to make (actual (non)“Feynman” rules from another course using Halzen and Martin)

To evaluate the matrix element $\mathcal{M}$ write down

1. the current associated with each of the fermion lines,
2. the propagators associated with each of the bosons,
3. the coupling strengths at each of the interaction vertices.

$\mathcal{M}$ is the product of these terms. Additionally you may have to calculate the four-momentum transfer q and impose four momentum conservation to evaluate the matrix element.

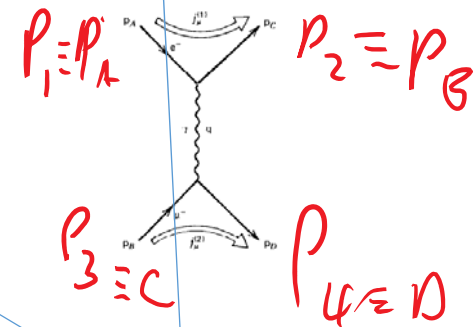
Matrix element:

$$jA^\mu = j_1 (1/q) j_2$$

Ignore why part of this slide for now just previous easy M tricks/Feynman rules schmules logic

$$-i\mathcal{M} = (ie(p_A + p_C)^\mu) \left(-i \frac{g_{\mu\nu}}{q^2} \right) (ie(p_B + p_D)^\nu).$$

This is regular matrix element!!! for t-channel
 $q = t$

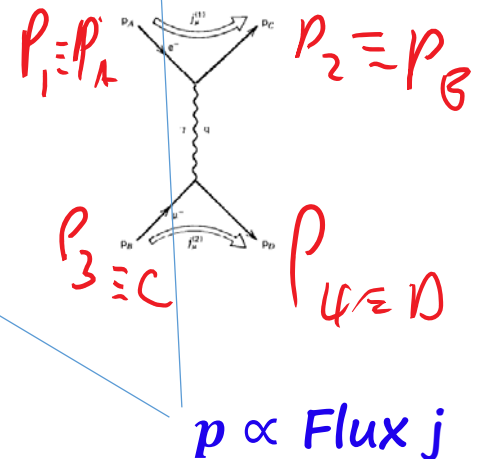


$p \propto \text{Flux } j$

Exercise

$$-i\mathcal{M} = \left(ie(p_A + p_C)^\mu \right) \left(-i \frac{g_{\mu\nu}}{q^2} \right) \left(ie(p_B + p_D)^\nu \right).$$

This is regular matrix element!!! for t-channel
 $q = t$



- show in High Energy limit s, t, u are interms of dot products of p_1, p_2, p_3, p_4
- Evaluate these products and write what the matrix element is proportional to in terms of s, t, u

Exercise

Toy ABC $\rightarrow$ Klein Gordon Equation

- Chapter 6 A BC Are Klein Gordon Fields
- K.G. Governs scalar fields $\phi(x)$
- Could describe interactions of pions (meaning pion-pion) pretty straightforwardly
- Can't find Chiral EFT treatment of pion-pion – some technical limitation – goldstone boson etc...

$$(\partial_\mu \partial^\mu + m^2) \phi = 0.$$

Preferred $u(p)$'s for

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▲
Peskin (Intro to QFT) is using the next symbols when discussing Dirac fields - $\sqrt{p\sigma}$ with $\sigma = (1, \sigma^1, \sigma^2, \sigma^3)$ (unit & Pauli).
8
▼
For example, he represents the Dirac spinor solution $u = (\sqrt{p\sigma}\xi^s, \sqrt{p\bar{\sigma}}\xi^s)$ for ξ^s a 2D basis and $\bar{\sigma} = (1, -\sigma^1, -\sigma^2, -\sigma^3)$
I dont understand the $\sqrt{p\sigma}$ symbol. Peskin says it's the matrix with the square root of the eigenvalues of $p\sigma$. But that doesn't make sence for two reasons,

▲
5
▼
Take a self-adjoint matrix A . There exists a unitary matrix U such that UAU^* is diagonal. Take the square root of every diagonal element in order to define $\sqrt{UAU^*}$ (now you are allowing for roots of negative numbers, so imaginary numbers as well). Then *rotate* the matrix back with U^* and set
$$\sqrt{A} := U^* \sqrt{UAU^*} U.$$

▲
3
▼
The form given in Peskin and Schroeder is useless except in the high energy and low energy limits and to remind the reader that taking the square of these matrices gives the result you would expect (e.g., $\sqrt{p \cdot \sigma^2} = p \cdot \sigma$). In practice a more useful form to use is,
$$\sqrt{p \cdot \sigma} \equiv \frac{E_p \pm m - \sigma \cdot \mathbf{p}}{\sqrt{2(E_p + m)}}$$
These can be derived from the other answer (though this is not a trivial exercise).

Halzen & Martin Spinless QED Preview

4.1 An "Electron" in an Electromagnetic Field A^μ

A free "spinless" electron satisfies the Klein-Gordon equation, (3.19):

$$(\partial_\mu \partial^\mu + m^2)\phi = 0.$$

In classical electrodynamics, the motion of a particle of charge $-e$ in an electromagnetic potential $A^\mu = (A^0, \mathbf{A})$ is obtained by the substitution

$$p^\mu \rightarrow p^\mu + eA^\mu \quad (4.1)$$

(see a standard text such as Goldstein or Jackson). The corresponding quantum-

$$i\partial^\mu \rightarrow i\partial^\mu + eA^\mu,$$

Klein-Gordon equation becomes

$$(\partial_\mu \partial^\mu + m^2)\phi = -V\phi,$$

where the electromagnetic perturbation is

$$V = -ie(\partial_\mu A^\mu + A^\mu \partial_\mu) - e^2 A^2.$$

$$\begin{aligned} T_{fi} &= -i \int \phi_f^*(x) V(x) \phi_i(x) d^4x \\ &= i \int \phi_f^* ie (A^\mu \partial_\mu + \partial_\mu A^\mu) \phi_i d^4x. \end{aligned}$$

$$T_{fi} = -i \int j_\mu^{fi} A^\mu d^4x,$$

$$j_\mu^{fi}(x) \equiv -ie(\phi_f^*(\partial_\mu \phi_i) - (\partial_\mu \phi_f^*)\phi_i)$$

$$\phi_i(x) = N_i e^{-ip_i \cdot x},$$

N_i is a normalization constant. Using a similar expression for ϕ_f

$$j_\mu^{fi} = -e N_i N_f (\underline{p_i + p_f})_\mu e^{i(p_f - p_i) \cdot x}.$$

This identification is done using Maxwell's equations

$$\square^2 A^\mu = j_{(2)}^\mu$$

(4.8)

$$A^\mu = -\frac{1}{q^2} j_{(2)}^\mu$$

(4.9)

$$T_{fi} = -i \int j_\mu^{(1)}(x) \left(-\frac{1}{q^2} \right) j_{(2)}^\mu(x) d^4x.$$

*M !!!! (no Feyn. rules) Griff ABC
only had middle factor 1/q^2 ...now p's too*

$$-i\mathcal{M} = (ie(\underline{p_A + p_C})^\mu) \left(-i \frac{g_{\mu\nu}}{q^2} \right) (ie(\underline{p_B + p_D})^\nu)$$

Map of Dirac Equation Complications

- γ Matrices
- Spinors (4D)
- Plain (plane) Wave Spinor Solutions u and v
- “Bilinear Covariants”
- Feynman rules – ~~Photon Spin!!!!~~ (not important for now, propagator today just $\propto 1/q^2$)
- Trace Theorems – Casimir’s Trick

Halzen & Martin Spinless QED Preview

$$\left\{ \underset{s}{2(p_1 \cdot p_2)} \underset{s}{(p_3 \cdot p_4)} + 2 \underset{u}{(p_2 \cdot p_3)} \underset{u}{(p_1 \cdot p_4)} - 2 \underset{T}{(p_1 \cdot p_3)} (Mc)^2 - 2 \underset{t}{(p_2 \cdot p_4)} (mc)^2 + 4(mMc^2)^2 \right\}$$

$$(s-u)^2 \quad \frac{s^2 + u^2 - 2su}{t}$$

$$p_1 \cdot p_2 + p_1 \cdot p_4 + p_3 \cdot p_2 + p_3 \cdot p_4$$

$T_{fi} = -i \int j_\mu^{(1)}(x) \left(-\frac{1}{q^2} \right) j_\mu^{(2)}(x) d^4x.$
M !!!! (no Feyn. rules) Griff ABC
only had middle factor $1/q^2$...now p's too
 $-i\mathcal{M} = (ie(\underline{p_A + p_C})^\mu) \left(-i \frac{g_{\mu\nu}}{q^2} \right) (ie(\underline{p_B + p_D})^\nu)$

First Reminder K.G. Notation

- Klein Gordon

$$\boxed{-\frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2} + \nabla^2 \psi = \left(\frac{mc}{\hbar} \right)^2 \psi}$$

$$\partial_\mu \partial^\mu = \partial^\mu \partial_\mu = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \equiv \square^2$$

$$(\partial_\mu \partial^\mu + m^2) \phi = 0.$$

- Leads to negative probabilities AND negative energy solutions
 - Negative energy solutions more OK $\rightarrow$ anti-particles Dirac, (Proca? spin 1) has too
 - Negative probability a problem, but not if we make $\Psi^* \Psi = \text{charge density}$ instead of number/counting probability

Exercise Pauli

- Warmup Exercise: Recall Pauli

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- Evaluate i,j component 2,1 of matrix O ($O_{ij} = O_{21}$) Enter into tophat
- $O = \sigma_1 (\sigma_2 \sigma_3 + \sigma_3 \sigma_2)$

Exercise Pauli

- Warmup Exercise: Recall Pauli

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- Evaluate i,j component 2,1 of matrix O ($O_{ij} = O_{21}$) Enter into tophat

$$O = \sigma_1 (\sigma_2 \sigma_3 + \sigma_3 \sigma_2)$$

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij}$$

Dirac IDEA : Factorize “KG”

- Put plain wave solution into KG

$$\left(\partial_\mu \partial^\mu + m^2 \right) \phi = 0.$$

$$p^\mu = \begin{pmatrix} p^0, & \vec{0} \end{pmatrix}$$

$$p_\mu p^\mu - m^2 c^2 = 0 \Rightarrow \left(p^0 \right)^2 - m^2 c^2 = 0 \Rightarrow \boxed{\left(p^0 + m \right) \left(p^0 - m \right) = 0}$$

This equation is easily satisfied by $p^0 = \pm mc$

- Idea for Dirac Eq mimic this idea

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- Trace Theorems – Casimir’s Trick

Factor full 4v

- For full four vectors enter vector "factors" β, γ

$$\underbrace{(p^\mu p_\mu - m^2 c^2)}_{(=0)} = (\beta^\kappa p_\kappa + mc)(\gamma^\lambda p_\lambda - mc)$$

$$\beta^\kappa \gamma^\lambda p_\kappa p_\lambda - m^2 c^2 - mc(\beta^\kappa p_\kappa - \gamma^\lambda p_\lambda)$$

if $\beta^\kappa = \gamma^\lambda$

- Then we want: $\gamma^\kappa \gamma^\lambda p_\kappa p_\lambda = p^\mu p_\mu$

- This will happen if $\gamma^\mu \gamma^\nu = g^{\mu\nu}$

- or if [since p_k 's are just numbers they can be interchanged and then

$$\gamma^\kappa \gamma^\lambda p_\kappa p_\lambda = \frac{1}{2} (\gamma^\mu \gamma^\lambda p_\kappa p_\lambda + \gamma^\mu \gamma^\lambda p_\lambda p_\kappa) = \frac{1}{2} \{ \gamma^\kappa, \gamma^\lambda \} p_\kappa p_\lambda$$

- so...] or if

$$\boxed{\{ \gamma^\mu, \gamma^\nu \} = 2g^{\mu\nu}}$$

now rewrite $\gamma^\lambda \gamma^\kappa p_\kappa p_\lambda$

- A key defining property of these gamma matrices

Map of Dirac Equation Complications

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4x4 $\rightarrow$ 4 Spinors

Now we have $p^\mu p_\mu - m^2 c^2 = (\gamma^\kappa p_\kappa + mc)(\gamma^\lambda p_\lambda - mc) = 0$ as desired.

Can choose either factor to define the Dirac equation.

By convention $\gamma^\mu p_\mu - mc = 0$. $\gamma^\mu \partial_\mu \psi + im\psi = 0$

Making the usual operator substitution $p_\mu \rightarrow i\hbar \partial_\mu$ and allowing the resulting expression to operate on the (now necessarily four component) wavefunction ψ yields the Dirac equation:

$$i\hbar \gamma^\mu \partial_\mu \psi - mc\psi = 0 \quad \psi = \begin{pmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \\ \psi_3 \end{pmatrix} \quad \text{Free particle solution}$$

It is important to note that while the Dirac spinor ψ is a four component object, is it NOT a four-vector. It's transformation properties are important
(it has more like 8 components actually because it's complex!)
could probably γ 's Ψ 's with real-only but it would be ~ 8 dimensional?

Plane Wave Spinor Solutions

- Plane Wave fn

$$\psi(\vec{r}, t) = a e^{-\frac{i}{\hbar}(Et - \vec{p} \cdot \vec{r})} \underbrace{u(E, \vec{p})}_{\text{4D spinor part}}$$

normalization factor: ignore for now

in Dirac eq:

$$\gamma^\mu p_\mu a e^{-\frac{i}{\hbar}(x \cdot p)} u - m c a e^{-\frac{i}{\hbar}(x \cdot p)} u = 0$$

$$\boxed{(\gamma^\mu p_\mu - mc)u = 0}$$

Expand this and then write in 2x2 matrix form: $\gamma^\mu p_\mu = \gamma^0 p_0 - \vec{\gamma} \cdot \vec{p}$

$$= \frac{E}{c} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \vec{p} \cdot \begin{pmatrix} 0 & \vec{\sigma} \\ -\vec{\sigma} & 0 \end{pmatrix} = \begin{pmatrix} E/c & -\vec{p} \cdot \vec{\sigma} \\ \vec{p} \cdot \vec{\sigma} & -E/c \end{pmatrix} \quad \left[\gamma^i = \begin{pmatrix} 0 & \vec{\sigma}_i \\ -\vec{\sigma}_i & 0 \end{pmatrix} \right]$$

SOLUTIONS: (Griff $\approx$ ~most commonly used—not only ones)

$$u_1 = N \begin{pmatrix} 1 \\ 0 \\ \frac{c(p_z)}{E + mc^2} \\ \frac{c(p_x + ip_y)}{E + mc^2} \end{pmatrix} \quad u_2 = N \begin{pmatrix} 0 \\ 1 \\ \frac{c(p_x - ip_y)}{E + mc^2} \\ \frac{c(-p_z)}{E + mc^2} \end{pmatrix}$$

$$E^2 = +\sqrt{m^2 c^4 + |\vec{p}|^2 c^2}$$

$$v^{(1)} = N \begin{pmatrix} \frac{c(p_x - ip_y)}{E + mc^2} \\ \frac{c(-p_z)}{E + mc^2} \\ 0 \\ 1 \end{pmatrix}, \quad v^{(2)} = -N \begin{pmatrix} \frac{c(p_z)}{E + mc^2} \\ \frac{c(p_x + ip_y)}{E + mc^2} \\ 1 \\ 0 \end{pmatrix}$$

$$\psi = a e^{-ip \cdot x / \hbar} u \quad (\text{particles}), \quad \psi = a e^{ip \cdot x / \hbar} v \quad (\text{antiparticles})$$

NORM

$$\boxed{N \equiv \sqrt{(E + mc^2)/c} \rightarrow u^\dagger u = 2|E|/c}$$

Comments about QM to Reltv. QM

- Dirac equation “great success” : spin “plops out”
 - b.c. of $> 4 \times 4$ dim requirement $\rightarrow 4\text{-D}^* (8)$ instead of $2\text{-D}^* (4) \rightarrow$ relativistic
- Note however this case is much more difficult than regular QM
 - There we just had direct/tensor product for spin* W.F.

$$\Psi(\vec{x}) \otimes \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

- They weren't “coupled” this way
- Coupling makes things much harder to understand
- E.g. before/reg. QM: what were 2-D spinors? Ans: “orientation” of spin
- now/Dirac: What are these 4-D spinors? Ans: Some weird mixture of momentum and spin (and anti-particleness?) (and ...?)
 - PHYSICAL CONTENT OBSCURED!!!
- But there is a way to see through this obscurity: form vectors w/ them

Exercise

- Take

Griff vs Other (e.g. Peskin) Expl of Spinors

- Definition of (4-D) Spinor is actually an object that LORENTZ transforms in a certain way

$$\psi' \rightarrow S \psi$$

- Peskin: (not very understandable :) define arbitrary transformation generators, show that 4-D rep is dirac spinors

- L/B : (similar) look at book ? Why do we need "4"-components ans: parity $\rightarrow$ extra 2 components encode parity

- Peskin: generators are not Hermitian leads to need of extra factor of γ^0

$$\psi^\dagger \not\rightarrow \psi^\dagger S^\dagger \rightarrow \bar{\psi} \equiv \psi^\dagger \gamma^0 \rightarrow \bar{\psi} S^{-1}$$

- Griff : derive solutions $\rightarrow$ empirically workout how $\Psi^* \Psi$ or $\Psi^\dagger \Psi$ or Prob density / current should transform

$$\bar{\psi} \psi = \psi^\dagger \gamma^0 \psi = |\psi_1|^2 + |\psi_2|^2 - |\psi_3|^2 - |\psi_4|^2$$

Bilinear Covariants

- 4-spinors not 4-v's, how to transform (need whole Dirac eq covariant/invariant) $i\hbar\gamma^\mu\partial_\mu\psi - mc\psi = 0$

- Can't transform $|\psi|^2$ properly though with just a single regular transformation matrix/operator—closest we can come is as follows—which we will eventually use but with a fix

$$\psi \rightarrow \psi' = S\psi \quad S = \begin{pmatrix} a_+ & a_- \sigma_1 \\ a_- \sigma_1 & a_+ \end{pmatrix}$$

Note that this is a 4x4 matrix. The σ_i are the 2x2 Pauli spin matrices (here we use σ_1 for the case of a boost along x).

works for ψ but not for $\psi^\dagger\psi$

with $a_\pm = \pm\sqrt{(\gamma \pm 1)/2}$ and $\gamma = (1 - \beta^2)^{-1/2}$

Btw normally (as w/ 2-D spin) we would create probability “ $\rho(x)$ ” = $|\psi|^2$ like this:

$$\psi^\dagger\psi = \begin{pmatrix} \psi_1^* & \psi_2^* & \psi_3^* & \psi_4^* \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix} = |\psi_1|^2 + |\psi_2|^2 + |\psi_3|^2 + |\psi_4|^2$$

But this is will not transform with the above matrix

As usual borrowing much from Kreiger Toronto

Transformation?

- S doesn't work by itself for $|\psi|^2$ because $S^\dagger S$ isn't 1 ($\mathbb{I}$)

How does $\psi^\dagger \psi$ transform under a Lorentz transformation (again along x)?

$$(\psi^\dagger \psi) \rightarrow (\psi^\dagger \psi)' = (\psi')^\dagger \psi' = \psi^\dagger S^\dagger S \psi \neq \psi^\dagger \psi \text{ since } S^\dagger S \neq 1$$

($\neq \mathbb{I}_{4 \times 4}$)

$$S^\dagger S = \begin{pmatrix} \boxed{a_+^2 + a_-^2} & 0 & 0 & 2a_+ a_- \\ \boxed{0} & \boxed{a_+^2 + a_-^2} & 2a_+ a_- & 0 \\ \boxed{0} & \boxed{2a_+ a_-} & \boxed{a_+^2 + a_-^2} & 0 \\ \boxed{2a_+ a_-} & 0 & 0 & \boxed{a_+^2 + a_-^2} \end{pmatrix}$$

$$= \begin{pmatrix} a_+^2 + a_-^2 & 2a_+ a_- \sigma_1 \\ 2a_+ a_- \sigma_1 & a_+^2 + a_-^2 \end{pmatrix} = \gamma \begin{pmatrix} 1 & -\beta \sigma_1 \\ -\beta \sigma_1 & 1 \end{pmatrix} \quad [\text{e.g. in } 2 \times 2 \text{ form}]$$

$$\text{since } \begin{cases} a_+^2 + a_-^2 = \frac{1}{2}(\gamma+1) + \frac{1}{2}(\gamma-1) = \gamma \\ 2a_+ a_- = -\sqrt{\gamma-1}\sqrt{\gamma+1} = -\sqrt{\gamma^2-1} = -\sqrt{\frac{1}{1-\beta^2}-1} = -\sqrt{\frac{1-1+\beta^2}{1-\beta^2}} = -\sqrt{\beta^2 \gamma^2} = -\beta \gamma \end{cases}$$

Problematic elements/factors (not 0 or 1) all have same form

This is a Griffiths problem solution, not gone over in class but for your info how the extra γ^0 inclusion happens

Fix

- Problem elements:

$$a_+^2 + a_-^2 \quad \text{or} \quad 2a_+a_- \quad (\text{from } a_+a_- + a_-a_+)$$

If the sign were reversed in each case, we would get

$$a_+^2 - a_-^2 = \frac{1}{2}(\gamma + 1) - \frac{1}{2}(\gamma - 1) = 1 \qquad a_+a_- - a_-a_+ = 0$$

So we can get what we need if we introduce a matrix that effects the required change of sign: (γ^0 will do the trick).

Just use result (as Griff does in text):

- So we change the prescription for finding “ $|\psi|^2$ ” to include an extra factor of γ^0 in a particular location

Define the adjoint spinor as $\bar{\psi} \equiv \psi^\dagger \gamma^0$

- so

- “ $|\psi|^2$ ” defined to be $\bar{\psi}\psi = \psi^\dagger \gamma^0 \psi = |\psi_1|^2 + |\psi_2|^2 - |\psi_3|^2 - |\psi_4|^2$ which is invariant

Matrix Elements of Operators

- Similarly any time we want to do a matrix element of an operator, sandwich between $\bar{\psi}$ and ψ

$$\langle \psi_a | \hat{O} | \psi_b \rangle \equiv \bar{\psi} \hat{O} \psi$$

Example?!? (sort of) Parity, the operator, in this matrix rep, it IS γ^0 !!!

$$\psi \rightarrow \psi' = \gamma^0 \psi$$

So our $\bar{\psi}\psi$ is an regular scalar (and invariant)

$$(\bar{\psi}\psi) \rightarrow (\bar{\psi}\psi)' = (\psi')^\dagger \gamma^0 \psi' = \psi^\dagger (\gamma^0)^\dagger \gamma^0 \gamma^0 \psi = \psi^\dagger \gamma^0 \psi = \bar{\psi}\psi$$

PS ψ is not an eigenvector of parity (and expectation value is not a number or a Lorentz invariant)

$$\bar{\psi} \gamma^0 \psi = \psi^\dagger \gamma^0 \gamma^0 \psi = \psi^\dagger \psi$$

Mechanically

- compare/view visually $\psi^\dagger \psi \rightarrow \psi^\dagger \gamma^0 \psi \equiv \bar{\psi} \psi$

$$(\bar{\psi}\psi) \rightarrow (\bar{\psi}\psi)' = (\psi^\dagger)' \gamma^0 \psi' = \psi^\dagger (\gamma^0)^\dagger \gamma^0 \gamma^0 \psi = \psi^\dagger \gamma^0 \psi = \bar{\psi}\psi$$

$$\begin{pmatrix} a^* & b^* & c^* & d^* \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$$

$$a^*a + b^*b$$

$$+ c^*c \dots$$

$$\psi^\dagger \psi$$

$$\begin{pmatrix} a^* & b^* & c^* & d^* \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$$

minus $\leftarrow$

$$a^*a + b^*b - c^*c - d^*d$$

$$\bar{\psi}\psi = \psi^\dagger \gamma^0 \psi \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Parity $\rightarrow$ Pseudoscalar/pseudovectors

- We can make a pseudoscalar with yet one more gamma matrix

$$\psi \rightarrow \psi' = \gamma^0 \psi$$

$$\bar{\psi} \gamma^5 \psi \text{ with } \gamma^5 \equiv i \gamma^0 \gamma^1 \gamma^2 \gamma^3$$

$$\gamma^5 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{in the usual 2x2 form}$$

Construct other constructs by sandwiching other gammas:

(1)	$\bar{\psi} \psi = \text{scalar}$	(one component)
(2)	$\bar{\psi} \gamma^5 \psi = \text{pseudoscalar}$	(one component)
(3)	$\bar{\psi} \gamma^\mu \psi = \text{vector}$	(four components)
(4)	$\bar{\psi} \gamma^\mu \gamma^5 \psi = \text{pseudovector}$	(four components)
(5)	$\bar{\psi} \sigma^{\mu\nu} \psi = \text{antisymmetric tensor}$	(six components)

We will be interested in γ^5 later (esp. for Weak int)

Where did Parity come from?

- Why did we suddenly insert parity into the conversation:
- See QFT Lancaster/Blundell chapter:
 - In many limits, the 4-D spinor becomes like a repeat of two 2-spinors e.g. $p \rightarrow \infty$ (1010) so you may ask :

36.3 Dirac and Weyl spinors

36.4 Basis states for superpositions

36.5 The non-relativistic limit of the Dirac equ...

Exercises

▼ 37 How to transform a spinor

37.1 Spinors aren't vectors

37.2 Rotating spinors

37.3 Boosting spinors

37.4 Why are there four components in the Di...

Exercises

37.4 Why are there four components in the Dirac equation?

The Dirac equation is written in terms of a four-component Dirac spinor ψ . These Dirac spinors have two degrees of freedom: the two components of ξ (or η). So why do we have four components in the Dirac equation?

We found that for the case of massless particles the Dirac spinors fell apart into two: left-handed Weyl spinors and right-handed Weyl spinors. Each of these spinors carried the same spin information in ξ (or η). The question is, then, why does the world need left- and right-handed fields. We will answer this question by asking what fundamental operation turns left-handed spinors into right-handed spinors? The answer is parity.

Many of the properties of fermions are invariant with respect to parity and so we require that Dirac particles have the same properties in a parity reversed world. (The weak interaction violates parity, but we ignore that for now.) The parity operation P does the following: it maps $x \rightarrow -x$ and $v \rightarrow -v$ and so it results in the changes to the generators:

$$P^{-1}KP = -K, \quad P^{-1}JP = +J \quad (37.20)$$

“the answer is parity ...”

How about vector

- All these examples are related to the probability density
- What is the vector there?

(1)	$\bar{\psi}\psi$ = scalar	(one component)
(2)	$\bar{\psi}\gamma^5\psi$ = pseudoscalar	(one component)
(3)	$\bar{\psi}\gamma^\mu\psi$ = vector	(four components)
(4)	$\bar{\psi}\gamma^\mu\gamma^5\psi$ = pseudovector	(four components)
(5)	$\bar{\psi}\sigma^{\mu\nu}\psi$ = antisymmetric tensor	(six components)

- LOOKS LIKE J
- It's just the j^μ !!!! The flux for the case of Dirac particles

- Recall what j_μ was from last time? and some other interesting things to say

Mixing of p/spin ? Jibberish?

- Considering that $u(p)/\psi$ are already mixture of spin and p then we need to make these extra sandwiches seems like these are jibberish!
- In fact I will call the kind of terms jibberish terms sometimes

$$-e\bar{\psi}_f\gamma_\mu\psi_i$$

$$-e\bar{u}_f\gamma_\mu u_i e^{i(p_f-p_i)\cdot x}$$

Flux j

but we will find there not as jibberish-y as they look

Think of the “vector” as flux

- Instead of jibberish : Flux = j_μ

$$\begin{aligned} j_\mu^{f,i} &= -e \bar{\psi}_f \gamma_\mu \psi_i \\ &= -e \bar{u}_f \gamma_\mu u_i e^{i(p_f - p_i) \cdot x} \end{aligned}$$

Flux j

Flux j

Also recall that generally a flux $j = \rho \vec{v} \propto \vec{p}$
for four- $j = (\rho, \vec{j})$

3- reasons for making this association:

Flux Reason 1): “Simple Electro-dynamics
(halzen and martin)

Halzen & Martin Spinless QED Preview

4.1 An "Electron" in an Electromagnetic Field A^μ

A free "spinless" electron satisfies the Klein-Gordon equation, (3.19):

$$(\partial_\mu \partial^\mu + m^2)\phi = 0.$$

In classical electrodynamics, the motion of a particle of charge $-e$ in an electromagnetic potential $A^\mu = (A^0, \mathbf{A})$ is obtained by the substitution

$$p^\mu \rightarrow p^\mu + eA^\mu \quad (4.1)$$

(see a standard text such as Goldstein or Jackson). The corresponding quantum-

$$i\partial^\mu \rightarrow i\partial^\mu + eA^\mu,$$

Klein-Gordon equation becomes

$$(\partial_\mu \partial^\mu + m^2)\phi = -V\phi,$$

where the electromagnetic perturbation is

$$V = -ie(\partial_\mu A^\mu + A^\mu \partial_\mu) - e^2 A^2.$$

$$\begin{aligned} T_{fi} &= -i \int \phi_f^*(x) V(x) \phi_i(x) d^4x \\ &= i \int \phi_f^* ie (A^\mu \partial_\mu + \partial_\mu A^\mu) \phi_i d^4x. \end{aligned}$$

$$T_{fi} = -i \int j_\mu^{fi} A^\mu d^4x,$$

$$j_\mu^{fi}(x) \equiv -ie(\phi_f^*(\partial_\mu \phi_i) - (\partial_\mu \phi_f^*)\phi_i)$$

$$\phi_i(x) = N_i e^{-ip_i \cdot x},$$

N_i is a normalization constant. Using a similar expression for ϕ_f

$$j_\mu^{fi} = -e N_i N_f (\underline{p_i + p_f})_\mu e^{i(p_f - p_i) \cdot x}.$$

This identification is done using Maxwell's equations

$$\square^2 A^\mu = j_{(2)}^\mu$$

(4.8)

$$A^\mu = -\frac{1}{q^2} j_{(2)}^\mu$$

(4.9)

$$T_{fi} = -i \int j_\mu^{(1)}(x) \left(-\frac{1}{q^2} \right) j_{(2)}^\mu(x) d^4x.$$

*M !!!! (no Feyn. rules) Griff ABC
only had middle factor 1/q^2 ...now p's too*

$$-i\mathcal{M} = (ie(\underline{p_A + p_C})^\mu) \left(-i \frac{g_{\mu\nu}}{q^2} \right) (ie(\underline{p_B + p_D})^\nu)$$

Cool : Maxwell's Eq

- Neat step in the Halzen and Martin

$$T_{fi} = -i \int j_\mu^{fi} A^\mu d^4x,$$

???

$$T_{fi} = -i \int j_\mu^{(1)}(x) \left(-\frac{1}{q^2} \right) j_\mu^{(2)}(x) d^4x.$$

- Discussed in Griff for photons
- (below may involve specific choice of gauge)
see also about flux / current j from last time

This identification is done using Maxwell's equations

$$\square^2 A^\mu = j_{(2)}^\mu$$

Maxwell?
e.s.
 $\nabla^2 V(x) = \rho \equiv j_\mu = 0$
 $\nabla \cdot \vec{E} = \rho$

$$A^\mu = -\frac{1}{q^2} j_{(2)}^\mu$$

(plane wave A^μ)

ie Fields/(potentials) due to sources

In this simpler way than Feynman rules, this is how the propagator appears!

Halzen & Martin Spinless QED Preview

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where the electromagnetic perturbation is

$$V = -ie(\partial_\mu A^\mu + A^\mu \partial_\mu) - e^2 A^2.$$

$$\begin{aligned} T_{fi} &= -i \int \phi_f^*(x) V(x) \phi_i(x) d^4x \\ &= i \int \phi_f^* ie (A^\mu \partial_\mu + \partial_\mu A^\mu) \phi_i d^4x. \end{aligned}$$

$$T_{fi} = -i \int j_\mu^{fi} A^\mu d^4x,$$

$$j_\mu^{fi}(x) \equiv -ie(\phi_f^*(\partial_\mu \phi_i) - (\partial_\mu \phi_f^*)\phi_i)$$

$$\phi_i(x) = N_i e^{-ip_i \cdot x},$$

N_i is a normalization constant. Using a similar expression for ϕ_f

$$j_\mu^{fi} = -e N_i N_f (\underline{p_i + p_f})_\mu e^{i(p_f - p_i) \cdot x}.$$

This identification is done using Maxwell's equations

$$\square^2 A^\mu = j_{(2)}^\mu$$

(4.8)

$$A^\mu = -\frac{1}{q^2} j_{(2)}^\mu$$

(4.9)

$$T_{fi} = -i \int j_\mu^{(1)}(x) \left(-\frac{1}{q^2} \right) j_{(2)}^\mu(x) d^4x.$$

M !!!! (no Feyn. rules)

Griff ABC

only had middle factor $1/q^2$...now p's too

$$-i\mathcal{M} = (ie(\underline{p_A + p_C})^\mu) \left(-i \frac{g_{\mu\nu}}{q^2} \right) (ie(\underline{p_B + p_D})^\nu)$$

Matrix Element again – focus

- Matrix element so much easier to make (actual (non)“Feynman” rules from another course using Halzen and Martin)

To evaluate the matrix element $\mathcal{M}$ write down

1. the current associated with each of the fermion lines,
2. the propagators associated with each of the bosons,
3. the coupling strengths at each of the interaction vertices.

$\mathcal{M}$ is the product of these terms. Additionally you may have to calculate the four-momentum transfer q and impose four momentum conservation to evaluate the matrix element.

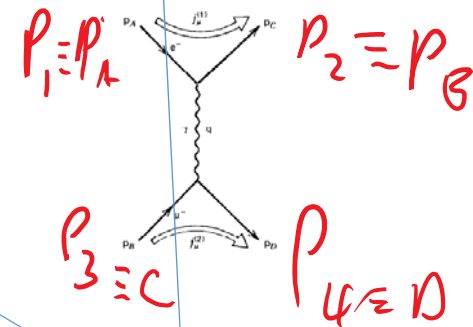
Matrix element:

$$jA^\mu = j_1 (1/q) j_2$$

Ignore why part of this slide for now just previous easy M tricks/Feynman rules schmoles logic

$$-i\mathcal{M} = (ie(p_A + p_C)^\mu) \left(-i \frac{g_{\mu\nu}}{q^2} \right) (ie(p_B + p_D)^\nu).$$

This is regular matrix element!!! for t-channel
 $q = t$



$p \propto \text{Flux } j$